Name:____________

A2 Chemistry Unit 4

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions your struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /80	Grade	Good Questions	Bad Questions	Reasons
2017					
2018					
2019					
2022					
2023					

Grade boundaries - Raw Score

Year	E	D	C	B	A	Max UMS
2017	18	25	32	39	47	67
2018	21	28	35	43	51	73
2019	21	29	37	45	53	69
2022	5	12	19	29	40	70
2023	11	19	27	35	44	69

Surname	Centre Number	Candidate Number
Other Names		2

GCE A LEVEL – NEW

1410U40-1

S17-1410U40-1

CHEMISTRY – A2 unit 4
Organic Chemistry and Analysis

MONDAY, 19 JUNE 2017 – MORNING

1 hour 45 minutes

	For Examiner's use only		
	Question	**Maximum Mark**	**Mark Awarded**
Section A	**1. to 8.**	**10**	
Section B	**9.**	**14**	
	10.	**14**	
	11.	**16**	
	12.	**12**	
	13.	**14**	
	Total	**80**	

1410U401
01

ADDITIONAL MATERIALS

In addition to this examination paper, you will need a:
- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions in the spaces provided.
Section B Answer **all** questions in the spaces provided.

Candidates are advised to allocate their time appropriately between **Section A (10 marks)** and **Section B (70 marks)**.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q.12***(a)*.

If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

JUN171410U40101

Examiner only

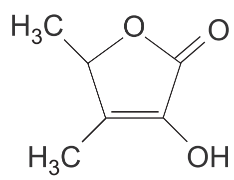
SECTION A

Answer **all** *questions in the spaces provided.*

1. State what is seen, if anything, when aqueous phenol is added to aqueous solutions of the following reagents. [2]

Reagent	Observation
iron(III) chloride	
sodium hydroxide	

2. Sotolone is one of the compounds that is responsible for the smell of raspberries.

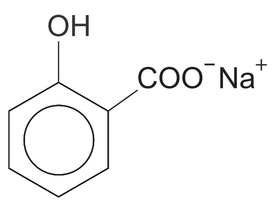


sotolone

Give the **empirical** formula of this compound. [1]

....

3. Complete the equation for the reaction of sodium 2-hydroxybenzoate with soda lime (NaOH), which, on strong heating, can give phenol as one of the products. [1]



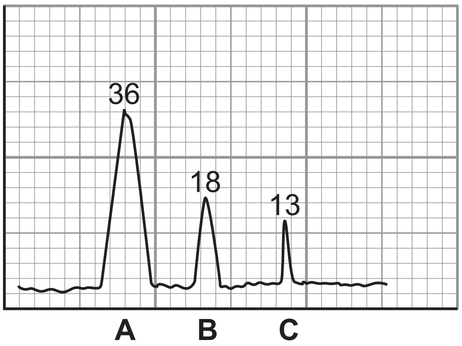
$+ \quad NaOH \longrightarrow$

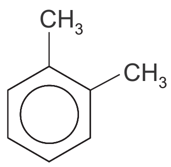
Examiner only

4. State the reagent used to produce phenylmethanol from benzaldehyde. [1]

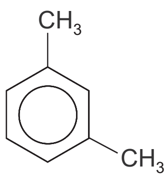
....

5. The gas chromatogram shows peaks for the products formed by the Friedel-Crafts alkylation of methylbenzene at 0 °C.

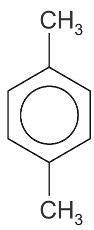




A



B



C

Use the data to calculate, to the appropriate number of significant figures, the percentage of 1,3-dimethylbenzene, which is present in the products. [1]

Percentage = ... %

6. Give the systematic name of a compound of formula $C_5H_{10}O_2$ that gives propan-1-ol as one of the products when it is warmed with aqueous sodium hydroxide. [1]

....

1410U401 03

Examiner only

7. Give the displayed formula of the organic compound obtained when cyclohexanol ($C_6H_{11}OH$) reacts with ethanoyl chloride. [1]

8. State the reagents necessary to convert hexane-1,6-diamine to hexane-1,6-dioic acid in two stages. [2]

$$H_2N(CH_2)_6NH_2 \xrightarrow{\textbf{A}} HO(CH_2)_6OH \xrightarrow{\textbf{B}} HOOC(CH_2)_4COOH$$

A ....

B ....

10

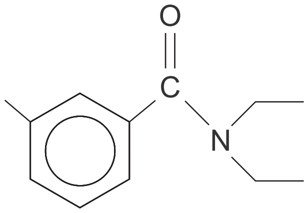
Examiner only

SECTION B

Answer **all** *questions in the spaces provided.*

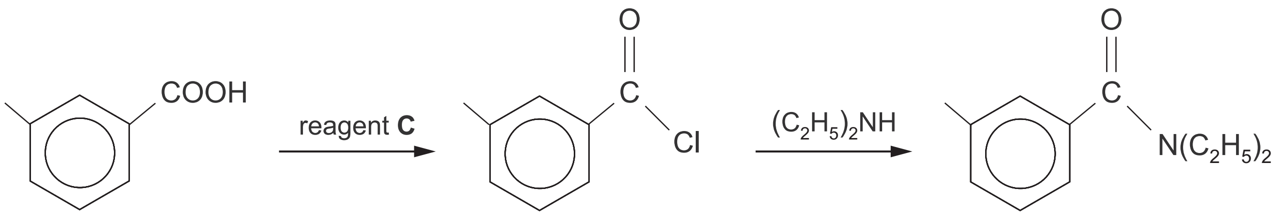
9. Insect repellents are materials that can be applied to the skin or to clothing in order to provide protection against biting insects.

(a) N,N-Diethyl-3-methylbenzamide, commonly known as DEET, has proved to be one of the most effective insect repellents.



DEET

(i) One method of preparing DEET is by the following reaction sequence.



Identify reagent **C** that can be used to convert 3-methylbenzoic acid to 3-methylbenzoyl chloride. [1]

....

(ii) DEET is a colourless liquid in white light.

Explain, in terms of the electromagnetic spectrum, why DEET is colourless. [1]

....

....

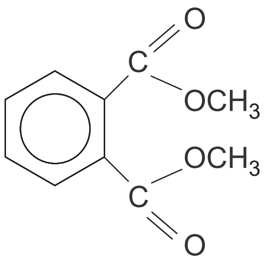
(iii) The mass spectrum of DEET shows a molecular ion at m/z 191. It also shows a prominent peak at m/z 91.

Suggest a formula for the fragment that is **lost** so that the remaining fragment has m/z 91. Show your working. [2]

Formula ....

Examiner only

(b) An alternative insect repellent is dimethyl benzene-1,2-dicarboxylate, DMP.



DMP

This can be made by the oxidation of 1,2-dimethylbenzene, followed by acidification of the resulting product and then esterification.

(i) State the oxidising agent used. [1]

....

(ii) Explain why it is necessary to acidify the product of this oxidation. [1]

....

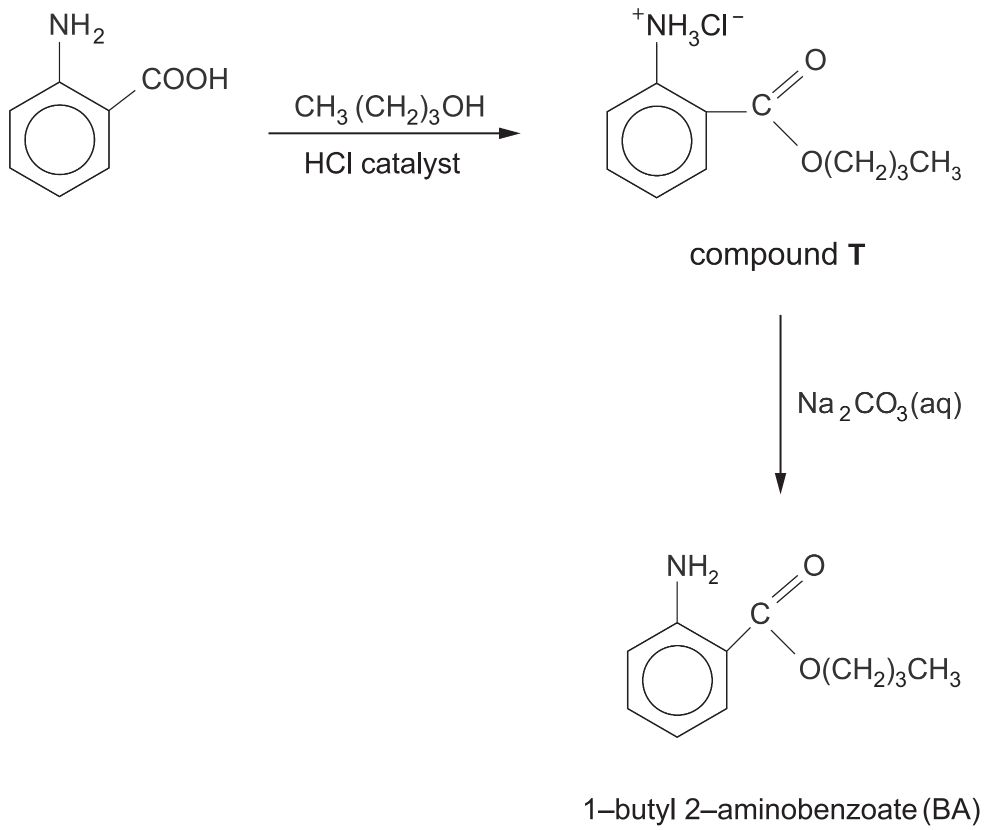
....

1410U401 07

Turn over.

Examiner only

(c) Another insect repellent is the 1-butyl ester of 2-aminobenzoic acid, known as BA. This can be produced from 2-aminobenzoic acid by esterification with butan-1-ol.



(i) Draw the zwitterion structure of 2-aminobenzoic acid. [1]

(ii) Explain why compound **T** is initially produced in this reaction. [1]

....

....

Examiner only

(iii) The ester BA is obtained from compound **T** by reacting it with sodium carbonate solution. It is then removed from the mixture by extracting it with the solvent ethoxyethane. After drying the ethoxyethane extract, the solvent is removed by distillation. Ethoxyethane has a boiling temperature of 35 °C and is very flammable.

I. Suggest the origin of the water removed during the drying process. [1]

....

....

II. Suggest how the ethoxyethane solution of BA could be safely heated to remove the solvent. [1]

....

....

(d) There is considerable interest in 'green' methods for the production of organic compounds. Recent studies have shown that BA can be produced from 2-aminobenzoic acid and butan-1-ol at 37 °C, using an enzyme as a catalyst.

(i) In a small scale experiment 4.0×10^{-4} mol of 2-aminobenzoic acid gave a 5 % yield of BA.

Calculate the number of moles of BA that were produced. [1]

n(BA) = ... mol

(ii) The esterification of 2-aminobenzoic acid is a reversible reaction and eventually the mixture will reach equilibrium. Use the information below to suggest why the addition of hexane to the stirred aqueous mixture of reactants increases the yield of the product BA. [3]

- hexane and the aqueous reaction mixture are immiscible
- 2-aminobenzoic acid is more soluble in water than in hexane
- BA is more soluble in hexane than in water

....

....

....

....

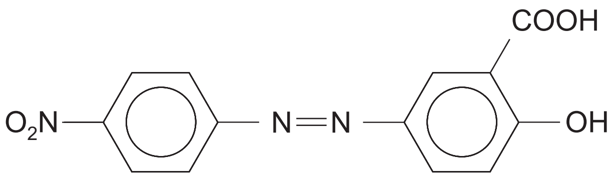
....

14

1410U401 09

Examiner only

10. *(a)* The azo dye Mordant Orange 1 has the formula shown below.



It is used as a pH indicator for acid-base titrations. At a pH of 10.1 the dye is yellow but at a pH of 12.1 it is orange-red.

(i) Give the displayed formulae of the two starting compounds that react, via a diazonium compound, to give Mordant Orange 1. [2]

(ii) The table shows the colour seen and the colour absorbed for a solution of Mordant Orange 1 as the pH increases.

Colour seen	yellow	orange-red
Colour absorbed	violet	greenish-blue
pH increasing →		

State how the wavelength of the light **absorbed** changes with pH, giving your reasoning. [1]

....

....

....

Examiner only

(iii) The visible spectrum of the dye at pH 10.1 shows that it has a maximum absorption at 385 nm.

Calculate the frequency of the radiation being absorbed at 385 nm. [2]

Frequency = ... Hz

(iv) Use your answer to (iii) to calculate the energy of the maximum absorption at 385 nm, giving your answer in kJ mol^{-1}. [3]

Energy = ... kJ mol^{-1}

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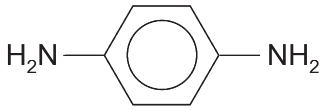
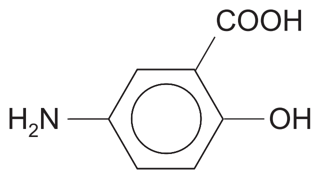
Examiner only

(b) Mordant Orange 1 can be reduced by the enzymes from certain bacteria to produce 1,4-diaminobenzene and 5-amino-2-hydroxybenzoic acid



1,4-diaminobenzene



5-amino-2-hydroxybenzoic acid

(i) 1,4-diaminobenzene can also be made from 4-nitrophenylamine.

State the reagent(s) necessary for this reaction. [1]

....

(ii) Use the displayed formula of 1,4-diaminobenzene to help you describe its ^1H high resolution NMR spectrum. Explain your reasoning.
Reference to the position of the signals is not required. [3]

....

....

....

....

....

....

....

(iii) Give the displayed formula of the organic compound formed when 5-amino-2-hydroxybenzoic acid reacts with aqueous sodium hydroxide in a 1:1 molar ratio. [1]

Examiner only

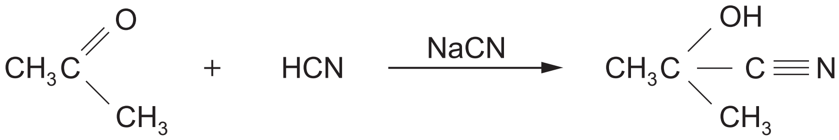
(iv) 5-Amino-2-hydroxybenzoic acid can be used in medicine to treat a number of conditions. It is believed to work by removing free radicals. One of these radicals is the hydroxyl radical. This can be produced by the homolytic fission of hydrogen peroxide, H_2O_2, where the O—O bond is broken.

Draw a dot and cross diagram of the hydroxyl radical, showing the outer electrons for each atom. [1]

14

Examiner only

11. *(a)* Propanone reacts with hydrogen cyanide (in the presence of sodium cyanide) to produce 2-hydroxy-2-methylpropanenitrile.



(i) Draw the mechanism for this reaction. Name the type of reaction mechanism occurring. [3]

Mechanism type ....

(ii) The following procedure was suggested to obtain 2-hydroxy-2-methylpropanenitrile.

mix the reactants, keep below 15 °C for 20 minutes with constant stirring → extract the reaction product with the solvent ethoxyethane → dry the ethoxyethane extract → remove the solvent → distil the product by vacuum distillation at 80 °C / 2 kPa pressure

The overall percentage yield was 55 %.

Suggest **two** stages in the procedure of extraction and distillation that could have resulted in this reduced yield. [2]

1. ....

....

2. ....

....

Examiner only

(iii) In a further experiment an excess of hydrogen cyanide reacted with 17.4 g of propanone (M_r 58.06) to produce 18.6 g of 2-hydroxy-2-methylpropanenitrile (M_r 85.07).

Calculate the percentage yield of 2-hydroxy-2-methylpropanenitrile. [3]

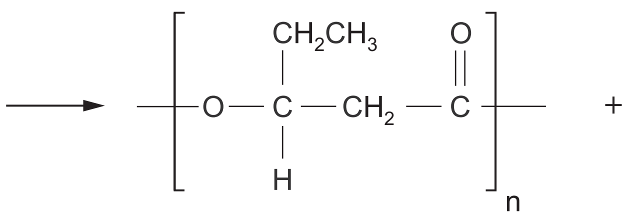
Percentage yield = ... %

(iv) The dehydration of 2-hydroxy-2-methylpropanenitrile produces 2-methylpropenenitrile, $H_2C{=}C(CH_3)CN$. This compound can undergo addition polymerisation giving poly(2-methylpropenenitrile).

Write the formula of the repeating section of this polymer. [1]

(b) There is considerable interest in biodegradable polymers. One of these is the polyester 'polyhydroxyvalerate' (PHV), which is produced from starch or glucose by using microorganisms. One simple chemical way of producing PHV is by the condensation polymerisation of 3-hydroxypentanoic acid.

Complete the equation below, which shows the structure of the repeating polymeric unit. [2]



3-hydroxypentanoic acid

Examiner only

(c) State **one** difference between condensation polymerisation and addition polymerisation. [1]

....

....

(d) Alcohols react with sodium to give hydrogen gas as one of the products.

$$2R\text{—}OH \quad + \quad 2Na \longrightarrow 2R\text{—}O^{-}Na^{+} \quad + \quad H_2$$

A student thought that the volume of hydrogen given off could be used to identify the alcohol.

In an experiment 0.900 g of the alcohol were dissolved in an inert solvent and an excess of sodium metal was added. 184 cm^3 of hydrogen were collected, measured at 298 K and 1 atm pressure.

(i) The reaction is exothermic. Suggest how the reaction mixture could be maintained at close to room temperature. [1]

....

....

(ii) Use the figures to calculate the relative molecular mass of the alcohol. [2]

M_r = ...

(iii) The alcohol used for this experiment gave a ketone on oxidation.

Give the displayed formula of the alcohol. [1]

16

Examiner only

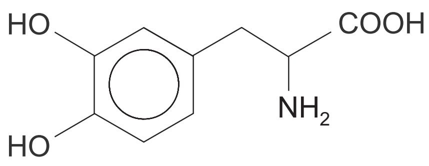
12. *(a)* Queen bees secrete a pheromone, compound **W**, which is used for a number of purposes, both inside and outside the hive.

Compound **W**
- is a straight chain aliphatic compound
- has a relative molecular mass of 184
- contains only carbon, hydrogen and oxygen
- contains 65.2% carbon and 26.1% oxygen by mass
- reacts with an alkaline solution of iodine to give a yellow solid
- produces effervescence when added to sodium hydrogencarbonate solution
- has a C=C double bond between carbons 2 and 3 and is an *E*-isomer
- produces an orange solid when reacted with 2,4-dinitrophenylhydrazine but does not give a silver mirror with Tollens reagent

Use all of this information to suggest a displayed formula for compound **W**. [6 QER]

Examiner only

(b) Broad bean plants produce the α-amino acid L-dihydroxyphenylalanine (L-DOPA), which has important uses in the treatment of Parkinson's disease.



L-DOPA

(i) Identify the chiral centre on the formula of L-DOPA by using an asterisk (*). [1]

(ii) A solution containing equimolar proportions of L-DOPA and its enantiomer D-DOPA is described as a racemic mixture.

Explain why this mixture has no apparent effect on the plane of plane polarised light. [2]

....

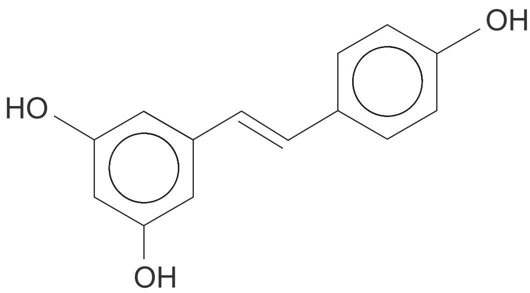
....

....

(iii) Write the displayed formula of the dipeptide formed from L-DOPA. [1]

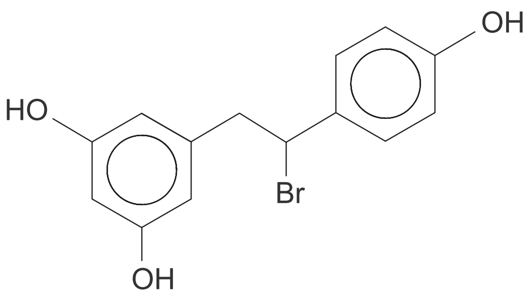
Examiner only

(c) Grape skins contain resveratrol.



resveratrol

(i) The addition of hydrogen bromide to resveratrol results in the compound whose formula is shown below.



Explain why this addition does not involve the aromatic rings. [1]

....

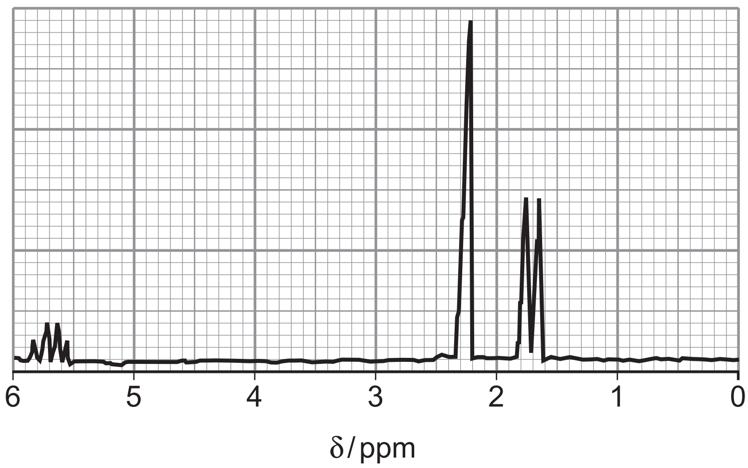
....

(ii) Suggest a displayed formula for a compound produced when excess aqueous bromine is added to resveratrol. [1]

12

Examiner only

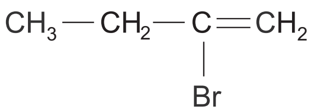
13. *(a)* The analysis of a compound shows that its formula is C_4H_7Br.
The high resolution ^1H NMR spectrum of the compound is shown below.



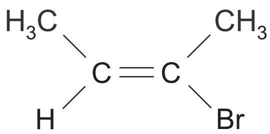
(i) A student suggested that the compound might be one of the following.

compound **L** $BrCH_2-CH_2-CH=CH_2$

compound **M**



compound **N**



Discuss the splitting pattern seen in the ^1H NMR spectrum to decide whether the compound is **L**, **M** or **N**. Give reasons for your answer. [4]

....

....

....

....

....

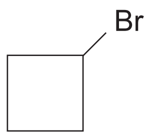
....

....

....

Examiner only

(ii) Another student suggested that the compound was bromocyclobutane.



I. Describe a chemical test that would show that the compound was either **L**, **M** or **N** and not bromocyclobutane. You should state the result of your test with each compound. [1]

....

....

....

II. Explain how the ^{13}C NMR spectrum would show that the compound was **L**, **M** or **N** and not bromocyclobutane. [2]

....

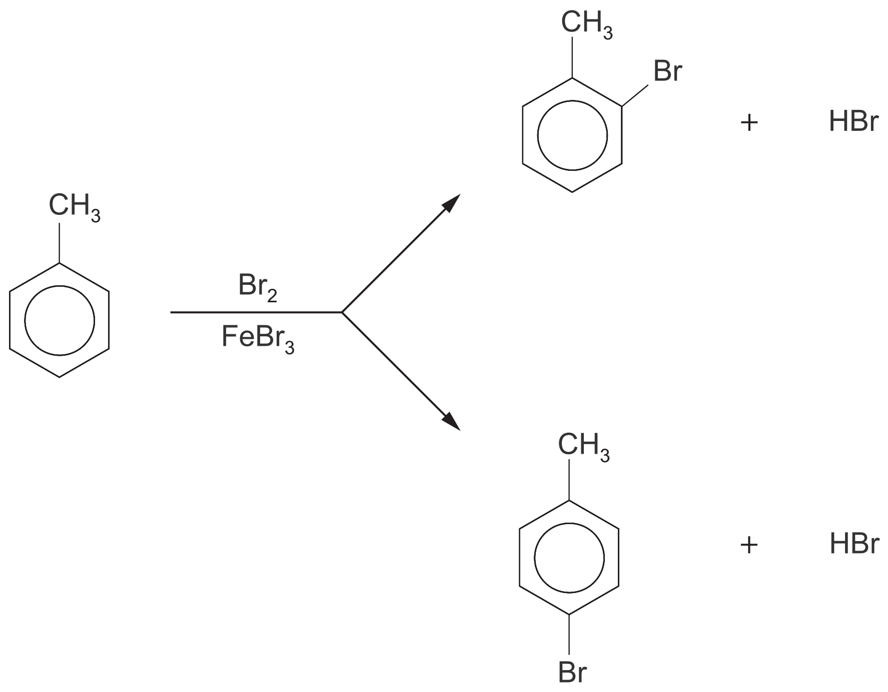
....

....

....

Examiner only

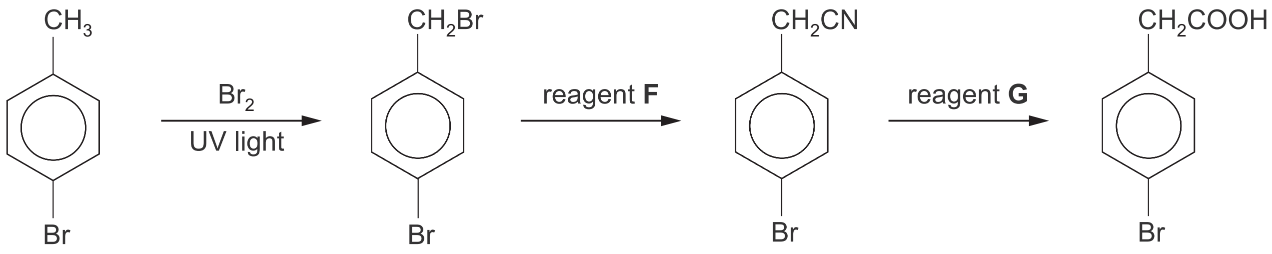
(b) The bromination of methylbenzene using electrophilic substitution gives a mixture of 2-bromomethylbenzene and 4-bromomethylbenzene.



Give the mechanism of the reaction that produces 4-bromomethylbenzene. [3]

Examiner only

(c) A reaction sequence to produce 4-bromophenylethanoic acid is shown below.



(i) The bromination of 4-bromomethylbenzene is a free radical process.

Give the equation for this reaction and then use the equation to calculate the minimum mass of bromine needed to convert 0.15 mol of 4-bromomethylbenzene to 4-bromo-1-(bromomethyl)benzene in the first stage of this reaction sequence. [2]

Mass of bromine = ... g

(ii) State the name of reagent **F**. [1]

....

(iii) State the name of reagent **G**. [1]

....

14

END OF PAPER

Surname	Centre Number	Candidate Number
Other Names		2

GCE A LEVEL

1410U40-1

S18-1410U40-1

CHEMISTRY – A2 unit 4
Organic Chemistry and Analysis

TUESDAY, 12 JUNE 2018 – AFTERNOON

1 hour 45 minutes

	For Examiner's use only		
	Question	**Maximum Mark**	**Mark Awarded**
Section A	**1. to 5.**	**10**	
Section B	**6.**	**14**	
	7.	**14**	
	8.	**14**	
	9.	**14**	
	10.	**14**	
	Total	**80**	

1410U401 01

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Section A Answer **all** questions in the spaces provided.
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Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q.9**(*a*).

If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

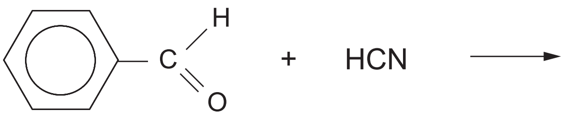
JUN181410U40101

Examiner only

SECTION A

Answer **all** *questions in the spaces provided.*

1. *(a)* Complete the equation below, which shows the reaction of benzaldehyde with hydrogen cyanide. [1]



(b) Give the **formula** of the nucleophile that takes part in the reaction shown in part *(a)*. [1]

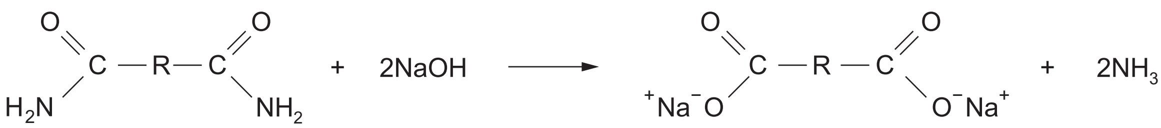
....

2. Decarboxylation occurs when a carboxylic acid is heated with sodalime.

State the name of the carboxylic acid that will produce propane when strongly heated with sodalime. [1]

....

3. Ammonia is produced when an amide is heated with aqueous sodium hydroxide.



amide **T**

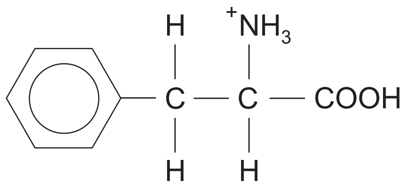
In an experiment 3.90 g of the aliphatic amide **T** gave 0.060 mol of ammonia.
Use this information to find the formula of the hydrocarbon group R, present in the amide **T**. [2]

Formula ....

 (1410U40-1)

Examiner only

4. *(a)* In acid solution the formula of the ion formed from phenylalanine is



Give the formula of the ion formed from phenylalanine in basic solution. [1]

(b) The melting temperatures of four compounds are shown in the table.

Compound	Melting temperature / °C
CH_3CH_2COOH	–21
$CH_3CH(OH)COOH$	18
$CH_3CHBrCOOH$	26
$CH_3CH(NH_2)COOH$	289

Explain why the melting temperature of 2-aminopropanoic acid is very much higher than the other acids in the table. [2]

....

....

....

Examiner only

5. On detonation, the explosive RDX produces carbon monoxide, steam and nitrogen.

$$C_3H_6N_6O_6(s) \longrightarrow 3CO(g) + 3H_2O(g) + 3N_2(g)$$

RDX
M_r 222

Calculate the mass of RDX that would give 1.00 m^3 of gaseous products measured at 100 °C and at 1 atm. Give your answer to an **appropriate** number of significant figures. [2]

(1 mol of any gas occupies a volume of 30.6 dm^3 under these conditions)

Mass of RDX = ... g

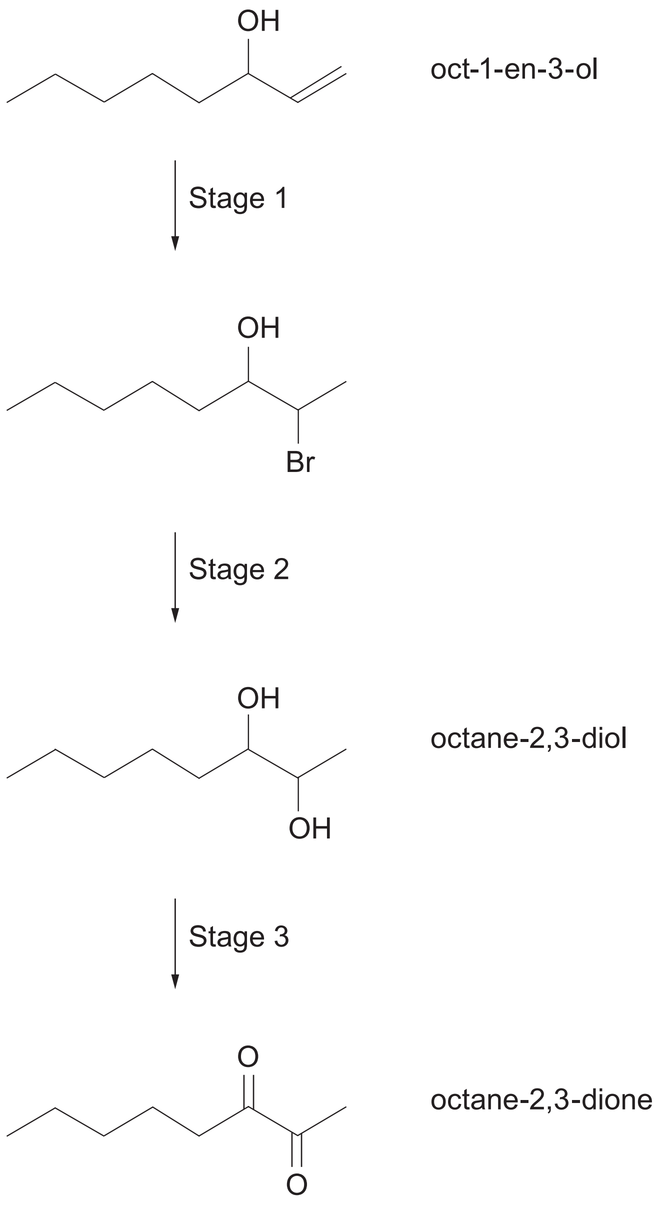
10

SECTION B

Answer **all** *questions in the spaces provided.*

6. *(a)* Oct-1-en-3-ol is a compound that attracts biting insects such as mosquitoes. It is contained in human breath and sweat and is also produced by some mushrooms.

A sequence of reactions starting with oct-1-en-3-ol is shown below.



Examiner only

(i) Give the molecular formula of oct-1-en-3-ol. [1]

....

(ii) State a reagent that can be used for Stage 2 and name the type of reaction mechanism taking place. [2]

Reagent(s) ....

Type of reaction mechanism ....

(iii) State a reagent that can be used for Stage 3. [1]

....

(b) An isomer of oct-1-en-3-ol was reacted with alkaline iodine and produced a yellow precipitate.

(i) Give the formula of the yellow precipitate. [1]

....

(ii) Suggest a structural formula for this isomer and identify the group that enables it to produce this yellow precipitate. [2]

Structural formula

Group ....

(c) Octane-2,3-dione is a pale yellow liquid and octane-2,3-diol is a colourless material.

(i) State a reagent that can be used to produce octane-2,3-diol from octane-2,3-dione. [1]

....

(ii) Suggest how the rate of the reaction in part (i) could be studied. [1]

....

....

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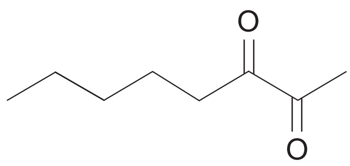
Examiner only

(d) The mass spectrum of octane-2,3-dione (M_r 142) shows fragmentation peaks at m/z 99, 71 and 43.



octane-2,3-dione

Identify the species responsible for these peaks. [2]

....

....

....

....

(e) Oct-1-en-3-ol is formed biochemically by the breakdown of linoleic acid.

$$CH_3(CH_2)_4CH{=}CHCH_2CH{=}CH(CH_2)_7COOH$$

linoleic acid

The carbon to carbon double bonds in linoleic acid are reduced by hydrogen and the resulting product can then be reduced by another suitable reducing agent to give a compound of formula $C_{18}H_{38}O$.

(i) Write the **skeletal** formula of this compound. [2]

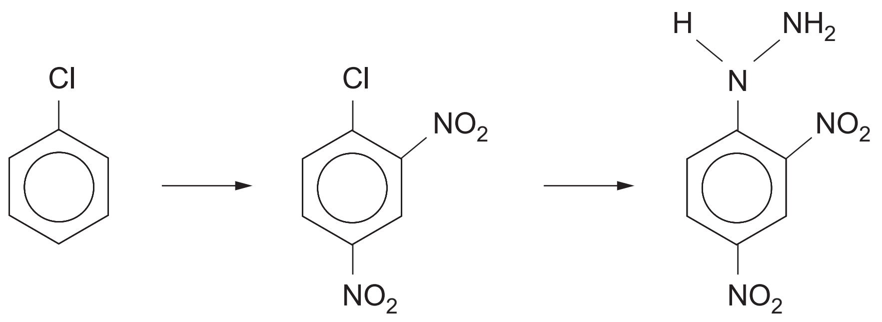
(ii) State the name of the reducing agent used in the second stage of the reduction. [1]

....

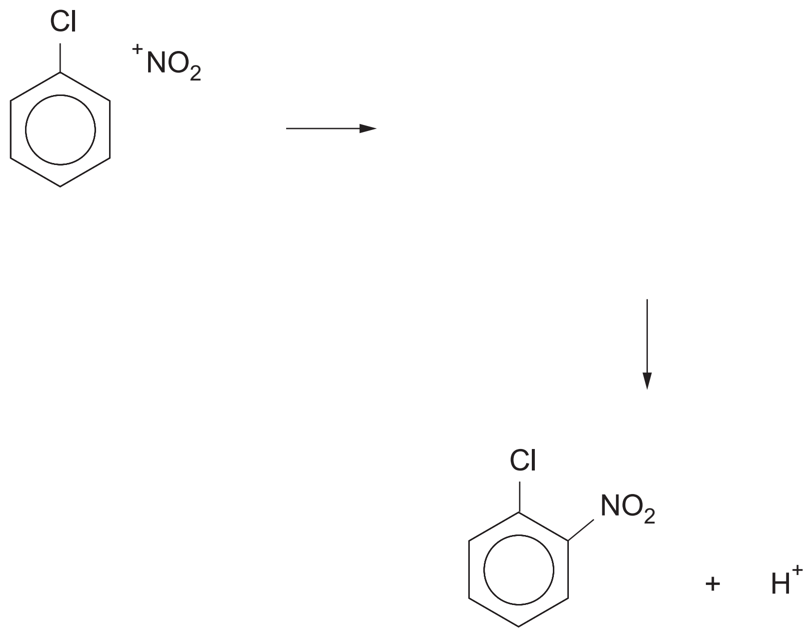
14

Examiner only

7. *(a)* 2,4-Dinitrophenylhydrazine (2,4-DNPH) can be made by the following route.



(i) Complete the mechanism for this reaction, which shows the nitration of chlorobenzene. You should use curly arrows and also show the structure of the intermediate. [2]



(ii) State the type of mechanism occurring in this nitration. [1]

....

(b) State the feature of 2,4-DNPH that enables it to react as a base. [1]

....

Examiner only

(c) 2,4-DNPH can be used as a reagent to identify aldehydes and ketones.

(i) State what is seen when 2,4-DNPH is used to identify an aldehyde or a ketone. [1]

....

(ii) Explain why 2,4-DNPH is an appropriate reagent to use in their identification. [1]

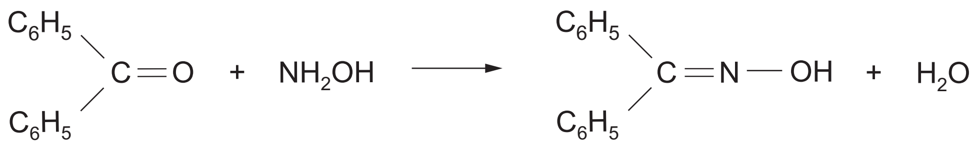
....

....

....

(d) Alternative reagents can be used for this identification. One of them is hydroxylamine, NH_2OH, which produces a white crystalline solid when it reacts with an aldehyde or ketone.

For example with diphenylmethanone



diphenylmethanone oxime
melting temperature 143 °C

In an experiment, impure diphenylmethanone oxime was recrystallised from flammable methanol (boiling temperature 65 °C).

Outline a **safe** procedure to obtain pure dry crystals of diphenylmethanone oxime using methanol as the solvent. You should assume that all the material present in the crude oxime is soluble in methanol. [5]

....

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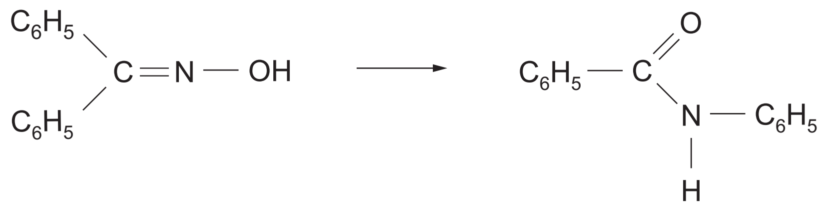
Examiner only

(e) Under suitable conditions diphenylmethanone oxime undergoes a rearrangement.



diphenylmethanone oxime

N-phenylbenzamide

Describe how the infrared absorption spectra of these two compounds would differ, giving the relevant bonds and their wavenumbers. You should comment on **both** compounds when considering differences. [3]

....

....

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Examiner only

8. *(a)* For a number of years CFCs were used in refrigeration and in air conditioning. The use of these compounds is in decline as they damage the ozone layer. A typical CFC is chlorotrifluoromethane.

Give an equation that shows the formation of a chlorine radical from chlorotrifluoromethane. [1]

....

(b) CFCs are being replaced by hydrofluorocarbons (HFCs) such as difluoromethane, which can be made by the reaction of dichloromethane with hydrogen fluoride in the presence of a catalyst.

$$CH_2Cl_2 + 2HF \longrightarrow CH_2F_2 + 2HCl$$

One problem with this reaction is that chlorofluoromethane is also produced.

The table below shows the percentage of the organic products formed (and unreacted dichloromethane) for various starting ratios of the reactants.

Mole ratio HF : CH_2Cl_2	Percentage composition of mixture / %		
	CH_2Cl_2	CH_2ClF	CH_2F_2
26 : 1	1.5	6.5	92.0
22 : 1	1.8	10.2	88.0
19 : 1	3.0	11.0	86.0
16 : 1	4.5	11.5	84.0

Examiner only

(i) Write a statement that links the mole ratio of HF : CH_2Cl_2 with the yield of difluoromethane. [1]

....

....

(ii) In an experiment using 0.040 mol of CH_2Cl_2, the yield of CH_2F_2 was 89.0%.
Calculate the mass of difluoromethane produced. [2]

Mass = ... g

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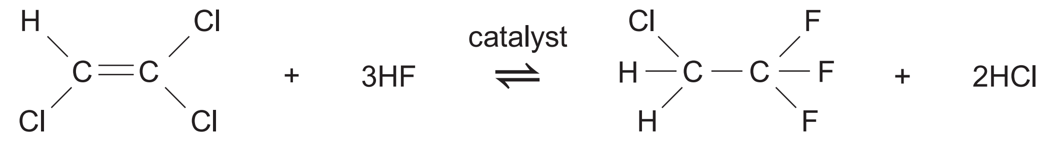
Examiner only

(c) 1,1,1,2-Tetrafluoroethane is an HFC that is made from trichloroethene and hydrogen fluoride in a two-stage vapour phase process.

Stage 1 – an exothermic process



Stage 2 – an endothermic process

(i) Suggest, giving a reason, whether a high or low pressure should be used in **stage 1**. [1]

....

....

(ii) Suggest, giving a reason, whether a high or low temperature should be used in **stage 2**. [1]

....

....

Examiner only

(iii) The boiling temperatures of the organic products formed after each stage are shown below.

Compound	Boiling temperature / °C
1,1,1-trifluoro-2-chloroethane	7
1,1,1,2-tetrafluoroethane	–26

The final product tends to be contaminated with some of the chlorocompound formed in the first stage.

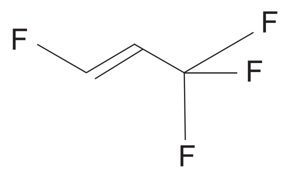
State the name of the method that can be used to separate these two compounds and suggest how this could be carried out in practice. [2]

....

....

....

(d) The use of HFCs is now declining as these compounds are contributing to global warming. Compounds such as HFOs (hydrofluoroolefins) are under development as suitable replacements. One such HFO is 1,3,3,3-tetrafluoropropene.



(i) Explain why 1,3,3,3-tetrafluoropropene exists as *E/Z* isomers. [1]

....

....

Examiner only

(ii) The addition of hydrogen bromide to 1,3,3,3-tetrafluoropropene results in two structural isomers, each of molecular formula $C_3H_3BrF_4$.

I. State the type of mechanism occurring in this reaction. [1]

....

II. These two compounds are formed in **approximately equal amounts** in this reaction.

Suggest why this ratio is different to that found in the reactions of other alkenes. [1]

....

....

....

....

(iii) Both the compounds formed in part (ii) will rotate the plane of plane polarised light.

Give the formula of **both** compounds indicating the chiral centre in both. [2]

(iv) Draw the two mirror image forms of the compound 1-bromo-1,2-difluoroethane, $CHBrFCH_2F$. [1]

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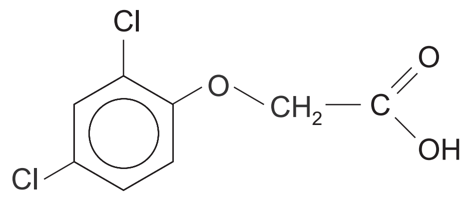
9. *(a)* A solid compound is believed to be the herbicide 2,4-dichlorophenoxyethanoic acid (2,4-D).



The following results were obtained when this compound was analysed.

- The percentage by mass of chlorine present was 32.1
- A solution of the compound reacted with sodium hydrogencarbonate to give a colourless gas
- 4.09 g of the compound was dissolved in a suitable solvent and the solution made up to 250 cm^3. A 25.0 cm^3 sample of this solution reacted with 24.70 cm^3 of aqueous sodium hydroxide of concentration 0.0750 mol dm^{-3}
- Refluxing a sample of the compound with aqueous sodium hydroxide did **not** show the presence of chloride ions in the resulting mixture
- The compound did **not** react with aqueous bromine to give a white precipitate or to decolourise the bromine
- The simplified ^{1}H NMR spectrum of the compound showed three separate peaks having the area ratio 1 : 2 : 3
- The ^{13}C NMR spectrum of the compound showed eight peaks

Examiner only

Use **all** the information to confirm that the compound is likely to be 2,4-D and suggest a simple laboratory method that would help to confirm that it is 2,4-D. [6 QER]

Examiner only

(b) An ester can be hydrolysed by heating it with aqueous sodium hydroxide solution.

(i) Give the equation for the hydrolysis of methyl ethanoate. [1]

(ii) The hydrolysis reaction can be used in a quantitative way to find the relative molecular mass of an ester.

A known mass of an ester is refluxed with a known volume of sodium hydroxide solution of concentration 1.0 mol dm^{-3} and the excess sodium hydroxide remaining after hydrolysis is determined by an acid-base titration.

The following results were obtained.

Mass of ester used = 1.76 g

Volume of aqueous sodium hydroxide added = 50.0 cm^3 (V_2)

Volume of aqueous sodium hydroxide remaining after hydrolysis = 30.0 cm^3 (V_1)

I. Use the formula below to calculate the M_r of the ester. [1]

$$\text{mass of ester} = \frac{M_r \text{ of the ester} \times (V_2 - V_1)}{1000}$$

M_r = ...

II. The ester reacts with Tollens' reagent to give a silver mirror. Use this information and the M_r of the ester found in part I to suggest a formula for the ester. Give a reason for your answer. [2]

....

....

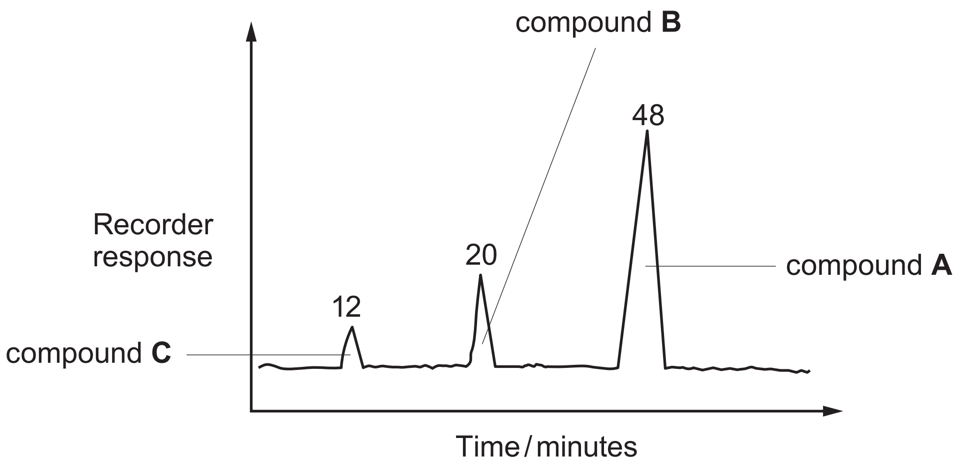
....

Examiner only

(iii) Esters of ethanoic acid with the formula $CH_3COOC_5H_{11}$ are used to give fruity flavours to confectionery.

I. The material used for this flavouring is often a mixture of esters having the formula above.

The diagram shows a gas chromatogram of a typical mixture.



Use the peak area data to calculate the percentage of compound **A** that is present in the mixture. [1]

Percentage = ... %

II. The peak areas shown give the molar percentage and in this case also the percentage by mass of each component present in the mixture.

Explain why the molar percentage of each compound **cannot** be used to find the percentage by mass of each component if the relative molecular mass of each component differs. [1]

....

....

....

Examiner only

(iv) The boiling temperatures of compounds **A**, **B** and **C** in part (iii) are shown in the table.

Compound	Name	R in $CH_3-C(=O)-O-R$	Boiling temperature / °C
A	2-methylbutyl ethanoate	$-CH_2CH(CH_3)CH_2CH_3$	134
B	3-methylbutyl ethanoate	$-CH_2CH_2CH(CH_3)CH_3$	142
C	pentyl ethanoate	$-CH_2CH_2CH_2CH_2CH_3$	147

Suggest a reason for the difference in boiling temperatures between compounds **A** and **C**. [1]

....

....

....

....

(v) Compound **A** is made by heating 2-methylbutan-1-ol and ethanoic acid in the presence of concentrated sulfuric acid.

State the name of another organic compound that could be formed if concentrated sulfuric acid reacted with 2-methylbutan-1-ol. [1]

....

14

Examiner only

10. *(a)* Butylamine can be made from propan-1-ol in a three-stage process.

Give the reagent used and the structural formula of the product for each stage. [3]

Stage	Reagent used	Structural formula of product
1		
2		
3		$CH_3CH_2CH_2CH_2NH_2$

(b) Below 10 °C, phenylamine reacts with nitric(III) acid, HNO_2, to give a benzenediazonium compound, which can react with phenol to give an azo dye.

(i) State how nitric(III) acid is prepared for use in this reaction. [1]

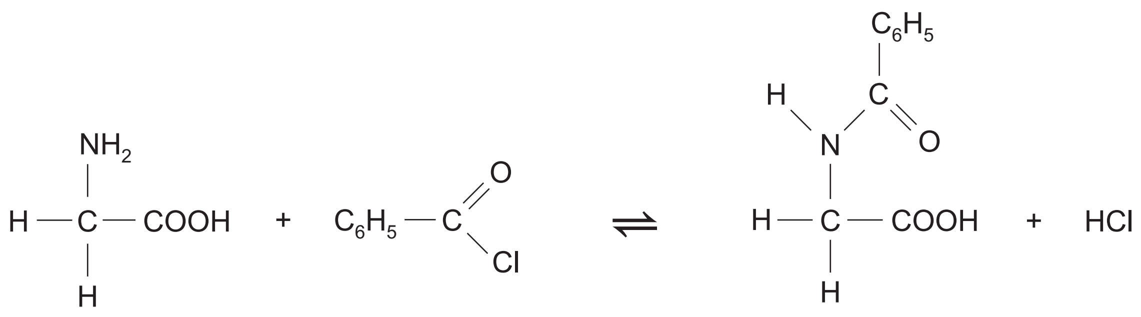
....

....

(ii) Give the equation for the reaction of benzenediazonium chloride with phenol to produce an azo dye. [1]

Examiner only

(c) Hippuric acid, so called because it is found in the urine of horses, can be made by reacting aminoethanoic acid and benzoyl chloride.



benzoyl chloride

hippuric acid

In an experiment the percentage yield was 90 %.

Benzoyl chloride is only slowly hydrolysed by alkalis. A teacher suggested the following idea for increasing the percentage yield.

'The experiment could be carried out in the presence of aqueous sodium hydroxide.'

Comment on the validity of this suggestion and any effect it may have on the yield. Explain your answer. [1]

....

....

Examiner only

(d) A straight chain α-amino acid of general formula $R—CH(NH_2)COOH$ reacts with nitric(III) acid to give nitrogen gas.

$$R—CH(NH_2)COOH \xrightarrow{HNO_2} R—CH(OH)COOH + N_2 + H_2O$$

In an experiment 7.87 g of the amino acid was dissolved in water and the solution made up to 500 cm^3. A 25.0 cm^3 sample of this solution was reacted with an excess of nitric(III) acid and gave 73.5 cm^3 of nitrogen measured at 298 K and 1 atm pressure.

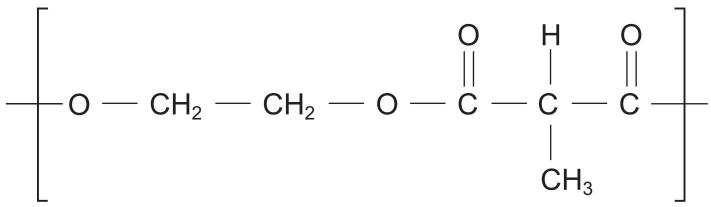
Calculate the molar mass of the amino acid and hence its structural formula. [4]

Molar mass = ...

Structural formula

Examiner only

(e) Polyesters can be made from a dicarboxylic acid and a diol, such as ethane-1,2-diol.



polyester **E**

(i) State the name of the dicarboxylic acid that reacts with ethane-1,2-diol to give polyester **E**. [1]

....

(ii) Complete the table to describe the high resolution ^{1}H NMR spectrum of polyester **E**. [3]

Protons	Splitting pattern	Relative peak area
$—CH_2—CH_2—$		
—CH—		
$—CH_3$		

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END OF PAPER

Surname	Centre Number	Candidate Number
Other Names		2

GCE A LEVEL

1410U40-1

S19-1410U40-1

TUESDAY, 11 JUNE 2019 – AFTERNOON

CHEMISTRY – A2 unit 4
Organic Chemistry and Analysis

1 hour 45 minutes

	For Examiner's use only		
	Question	**Maximum Mark**	**Mark Awarded**
Section A	**1. to 7.**	**10**	
Section B	**8.**	**13**	
	9.	**15**	
	10.	**13**	
	11.	**13**	
	12.	**16**	
	Total	**80**	

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ADDITIONAL MATERIALS

In addition to this examination paper, you will need a:
- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions in the spaces provided.
Section B Answer **all** questions in the spaces provided.

Candidates are advised to allocate their time appropriately between **Section A (10 marks)** and **Section B (70 marks)**.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q.11**_(a)_(i).

If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

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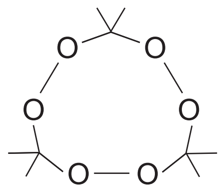
Examiner only

SECTION A

Answer **all** *questions in the spaces provided.*

1. The skeletal formula of the compound TATP is shown below.



Give the **molecular** formula of this compound. [1]

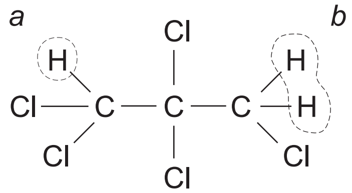
....

2. Alkanes are produced when carboxylic acids are strongly heated with sodalime.

State the name of a carboxylic acid that will produce propane, when heated in this way. [1]

....

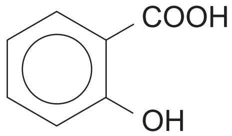
3. Complete the table below, which describes the ^1H NMR spectrum of 1,1,2,2,3-pentachloropropane. [2]

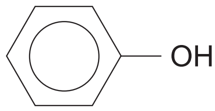


Proton(s)	Splitting pattern	Relative peak area ratio
a		
b		

Examiner only

4. Using the letters provided, place these three compounds in order of **increasing** acidity. [1]

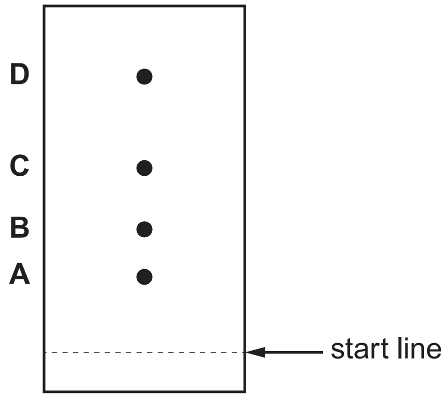




A **B** **C**

least acidic .... most acidic

5. A student produced the following chromatogram of a mixture of four amino acids.
The student forgot to mark the position of the solvent front.



The R_f values of the four amino acids are as follows.

Amino acid	R_f value
alanine	0.59
glycine	0.42
leucine	0.88
serine	0.36

State and explain which of the spots is likely to be alanine. [1]

....

....

....

Examiner only

6. Describe a simple chemical test that will distinguish between pentan-2-one and 1-phenylbutanone. [2]

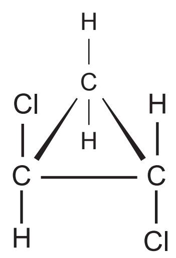
pentan-2-one 1-phenylbutanone

Reagent(s) used ....

Observations with both compounds ....

....

7. The displayed formula of 1,2-dichlorocyclopropane is given below.



(a) Draw the displayed formula of another structural isomer of formula $C_3H_4Cl_2$. [1]

(b) Draw the displayed formula of a stereoisomer of 1,2-dichlorocyclopropane. [1]

10

Examiner only

SECTION B

*Answer **all** questions in the spaces provided.*

8. *(a)* Chloroethanoic acid is produced from ethanoic acid by passing chlorine gas into it under suitable conditions.

$$CH_3COOH + Cl_2 \longrightarrow ClCH_2COOH + HCl$$

M_r 60.0 $\qquad\qquad$ M_r 94.5

(i) This reaction involves radicals. In the first stage chlorine radicals are formed. They then attack the ethanoic acid molecule producing a molecule of hydrogen chloride and a new radical.

Suggest the formula of the new radical formed from the acid during this reaction. [1]

....

(ii) I. In an experiment 89.0 g of ethanoic acid reacted with chlorine.

Calculate the increase in mass that occurs if chloroethanoic acid is the only organic product. [2]

Increase in mass = ... g

II. Chloroethanoic acid reacts with ammonia producing aminoethanoic acid.

$$CH_3COOH \longrightarrow ClCH_2COOH \longrightarrow CH_2(NH_2)COOH$$

In the experiment described in I, 89.0 g of ethanoic acid then gave 49.2 g of aminoethanoic acid (M_r 75.0).

Calculate the percentage yield of aminoethanoic acid. Give your answer to an **appropriate** number of significant figures. [2]

Percentage yield = ... %

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Examiner only

III. Aminoethanoic acid is isolated from the solution of the products by adding methanol to the liquid. The acid is precipitated as a white solid.

Suggest why aminoethanoic acid is insoluble in methanol. [2]

....

....

....

....

(b) Explain why an aqueous solution of aminoethanoic acid does not affect the plane of plane polarised light. [1]

....

....

(c) Draw the structure of the dipeptide formed from two molecules of aminoethanoic acid. [1]

Examiner only

(d) Aminoethanoic acid reacts with nitric(III) acid to produce nitrogen gas.

$$CH_2(NH_2)COOH \quad + \quad HNO_2 \longrightarrow CH_2(OH)COOH \quad + \quad N_2 \quad + \quad H_2O$$

Calculate the maximum volume of nitrogen gas produced at 100 °C and 98 kPa pressure when 0.300 mol of aminoethanoic acid reacts in this way. Give your answer in dm^3. [3]

Volume produced = ... dm^3

(e) Hydroxyethanoic acid, $CH_2(OH)COOH$, can also be prepared by reacting chloroethanoic acid with aqueous sodium hydroxide.

Explain why the reaction mixture needs to be acidified to produce hydroxyethanoic acid. [1]

....

....

....

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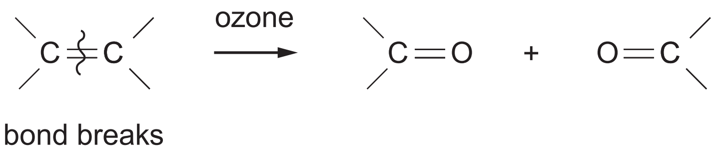
Examiner only

9. *(a)* Alkenes react with ozone to give an intermediate compound, which can then react further to give aldehydes or ketones.



1 mol of an alkene **T** reacts with ozone to give 2 mol of the same carbonyl compound, **U**.

(i) Explain what can be deduced about the structural formula of alkene **T**. [1]

....

....

(ii) Carbonyl compound **U** reacted with 2,4-dinitrophenylhydrazine to produce a solid derivative.

State the colour of this solid derivative. [1]

....

(iii) I. The ^{1}H NMR spectrum of compound **U** did **not** show a chemical shift at 9.8 δ.

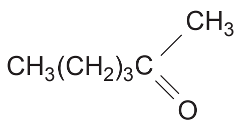
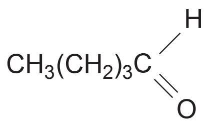
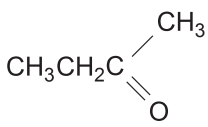
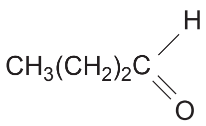
State what can be deduced from this statement. [1]

....

....

Examiner only

II. The melting temperatures of the 2,4-dinitrophenylhydrazine derivatives can be used to identify the original carbonyl compound. The melting temperatures of some of these derivatives are shown in the table.

Carbonyl compound	Melting temperature of derivative / °C
	106
	108
	115
	126

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The 2,4-dinitrophenylhydrazine derivative of compound **U** was slightly impure and melted at a temperature of 110-113 °C.

Use the information given to deduce the formula of compound **U**. Explain your reasoning. [3]

....

....

....

....

....

....

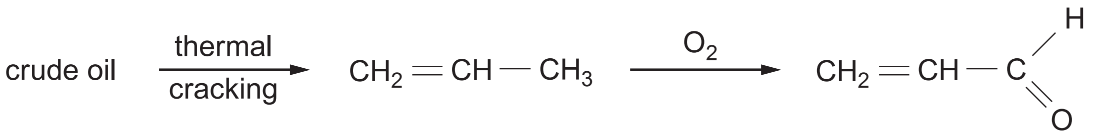
(iv) Use your answer to part (iii) and other relevant information given in the question to give the structure of alkene **T**. Name alkene **T**. [2]

Name ....

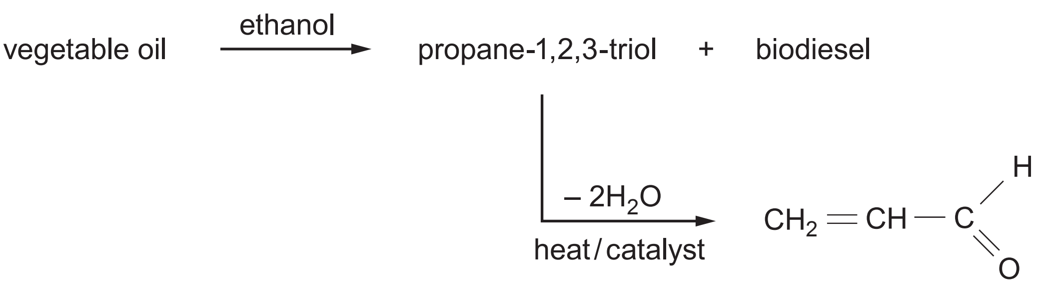
Examiner only

(b) Propenal, $CH_2{=}CH{-}CHO$ is made by two main methods.

Method 1 From crude oil



Method 2 From vegetable oil



(i) Suggest **one** reason why the general public would probably favour the production of propenal by Method 2. [1]

....

....

(ii) If you were considering which of these two methods to choose to produce propenal, state **two** other factors, apart from cost, that you would need to know to be able to come to a decision. [2]

Factor 1 ....

....

Factor 2 ....

....

Examiner only

(iii) The densities of biodiesel and propane-1,2,3-triol are shown in the table. These two liquids are immiscible (insoluble in each other).

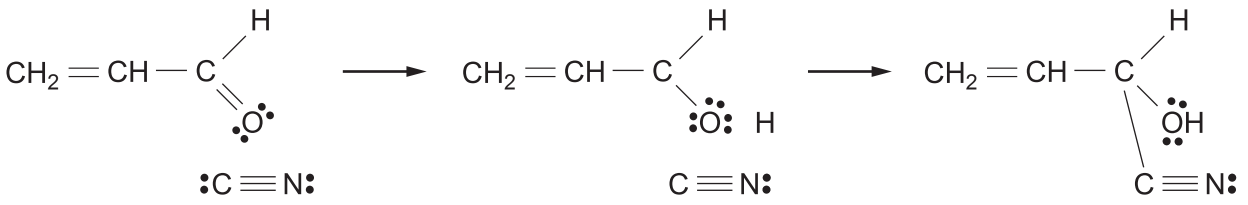
Liquid	Density / $g\,cm^{-3}$
biodiesel	0.85
propane-1,2,3-triol	1.26

Sketch and name a piece of apparatus that could be used to separate a mixture containing $25\,cm^3$ of each liquid. The two liquids should also be labelled. [2]

Name of apparatus ....

(iv) Propenal reacts with hydrogen cyanide by a nucleophilic addition reaction.

Complete the mechanism for this reaction using curly arrows and partial/complete charges as appropriate. [2]



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  Turn over.

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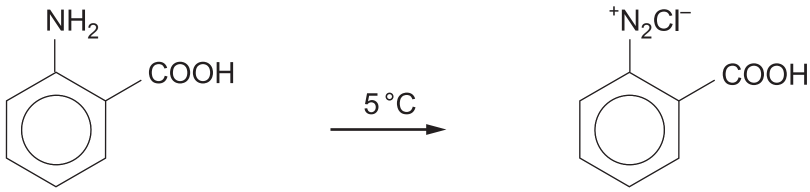
10. *(a)* The acid-base indicator methyl red can be made from 2-aminobenzenecarboxylic acid.

(i) In the first stage of the reaction, a benzenediazonium compound is made from this acid.

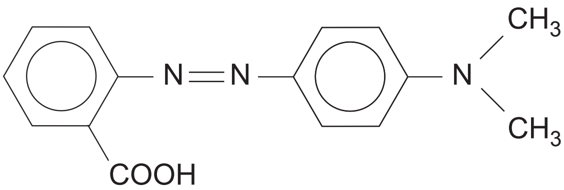


2-aminobenzenecarboxylic acid

State the reagent(s) necessary to produce this diazonium compound from the acid. [1]

....

(ii) The benzenediazonium compound is then coupled with an amine to make methyl red.



methyl red

Draw the structure of the amine used. [1]

(iii) The visible spectrum of this azo dye at a certain pH shows an absorption maximum at 410 nm.

Calculate the frequency that corresponds to this wavelength. [2]

Frequency = ... Hz

Examiner only

(b) Some azo dyes are permitted for use as food colours but their concentration is strictly controlled. The concentration present can be found by colorimetry. The absorption of a series of standard solutions is measured and a straight line calibration graph is drawn.

(i) The molar mass of an azo dye is 272 g mol^{-1}.

Calculate the mass of the azo dye needed to prepare 250 cm^3 of a standard solution of concentration 0.0075 mol dm^{-3}. [2]

Mass required = ... g

(ii) Absorption (*A*) and concentration (*c*) are related by the equation

$$A = k\,c \quad \text{where } k \text{ is a constant}$$

The calibration graph shows that an absorption of 1.44 is measured for a solution of concentration 0.0096 mol dm^{-3}.

Calculate the concentration of a solution that gave an absorption of 1.03. [2]

Concentration = ... mol dm^{-3}

(c) (i) When ethanoic acid is heated with urea, NH_2CONH_2, the products are ethanamide, carbon dioxide and ammonia.

Give the equation for this reaction. [1]

....

(ii) Suggest why an amide can act as a base. [1]

....

....

Examiner only

(iii) When ethanamide is heated with a dehydrating agent the organic product is ethanenitrile.

Use the **Data Booklet** to state and explain how characteristic infrared absorption values change during the reaction. In your answer you should relate the relevant bonds to their corresponding absorption frequencies. [2]

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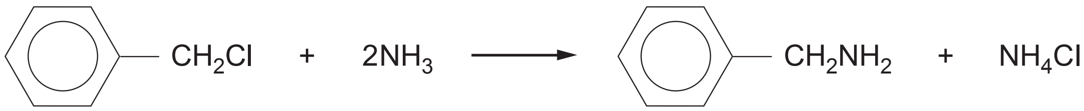
(iv) Some amines can be made by the reaction of a chlorine-containing compound and ammonia. For example (chloromethyl)benzene reacts with ammonia to give phenylmethylamine.



Suggest why phenylamine cannot easily be made by this method from chlorobenzene and ammonia. [1]

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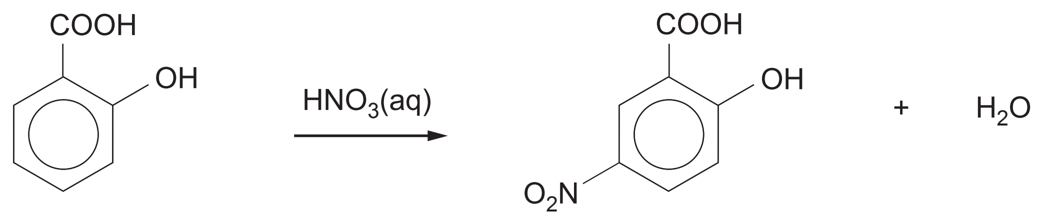
Examiner only

11. *(a)* 2-Hydroxy-5-nitrobenzenecarboxylic acid is made by reacting 2-hydroxybenzenecarboxylic acid and aqueous nitric acid.



(i) A basic outline of the method is given below.

Add 4.00 g of 2-hydroxybenzenecarboxylic acid to 100 cm^3 of nitric acid of concentration 2 mol dm^{-3}. Heat the mixture to 60 °C for 10 minutes. As the reaction proceeds brown fumes containing toxic nitrogen dioxide are produced. Add the products to 300 cm^3 of cold water. A yellow solution and white crystals of 2-hydroxy-5-nitrobenzenecarboxylic acid are obtained.

Use this outline method to produce a more detailed method suitable for others to use to be able to produce dry white crystals of the acid. As part of your answer you should give the type and size of apparatus used and mention any necessary health and safety factors.

Details of subsequent recrystallisation are **not** required. [6 QER]

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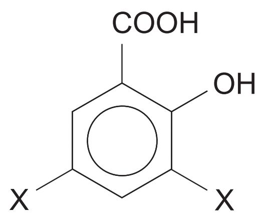
(ii) In this experiment the percentage yield of dry 2-hydroxy-5-nitrobenzenecarboxylic acid (M_r 183) was 41 %.

Calculate the mass produced based on the original mass of the 2-hydroxybenzenecarboxylic acid (M_r 138). [3]

Mass = ... g

(iii) Several other products were obtained during this reaction. One of these products is compound **J**.

Spectroscopy and analysis show that compound **J** is a substituted hydroxybenzenecarboxylic acid of M_r 228. Each molecule contains seven oxygen atoms.



compound **J**

Use the information given to suggest a structure for compound **J**. Show your reasoning. [2]

....

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Examiner only

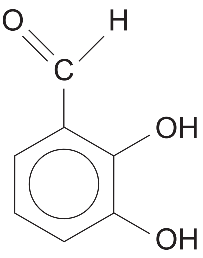
(iv) The yield of 2-hydroxy-5-nitrobenzenecarboxylic acid in this experiment is low.

Suggest how the experiment could be modified to increase the yield of the required product. [1]

....

....

(b) 2,3-Dihydroxybenzaldehyde has the same molecular formula as 2-hydroxybenzenecarboxylic acid.



Suggest a simple reaction that will give a positive result for 2-hydroxybenzenecarboxylic acid but **not** for 2,3-dihydroxybenzaldehyde. You should give the reagent(s) used and the result of the test with the acid. [1]

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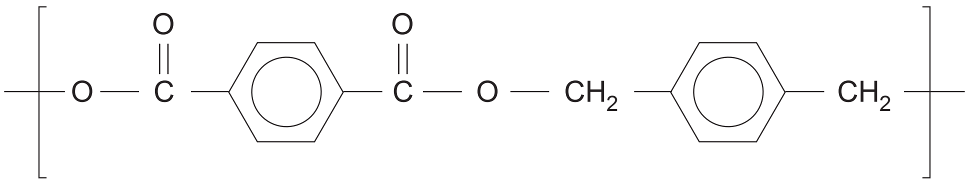
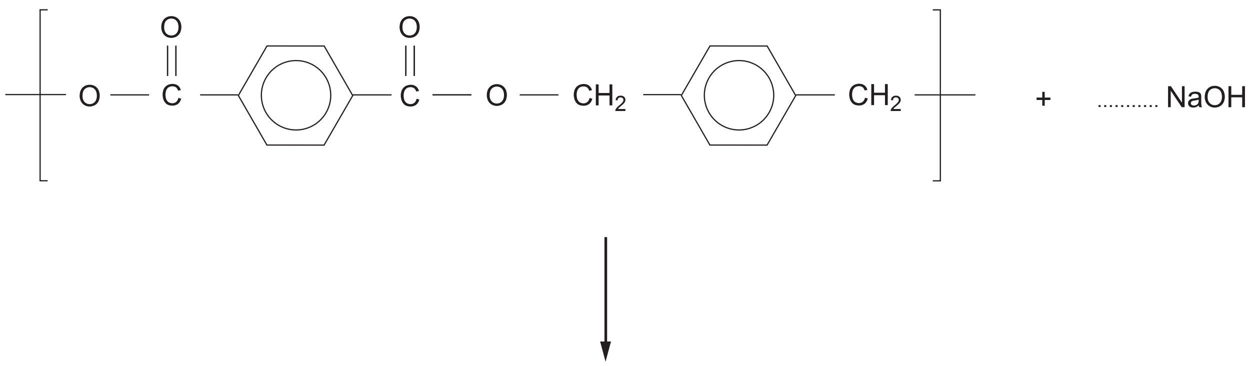
12. *(a)* A student was given a sample of a polyester **P**. He was told that the repeating unit was as follows.



(i) Draw a **circle** around an ester linkage in the repeating unit. [1]

(ii) Polyesters such as compound **P** can be hydrolysed by heating with aqueous sodium hydroxide.

Complete the balanced equation for this hydrolysis. [2]



(iii) The formation of polyester **P** is an example of condensation polymerisation.

State how this type of polymerisation differs from addition polymerisation. Refer to both types in your answer. [1]

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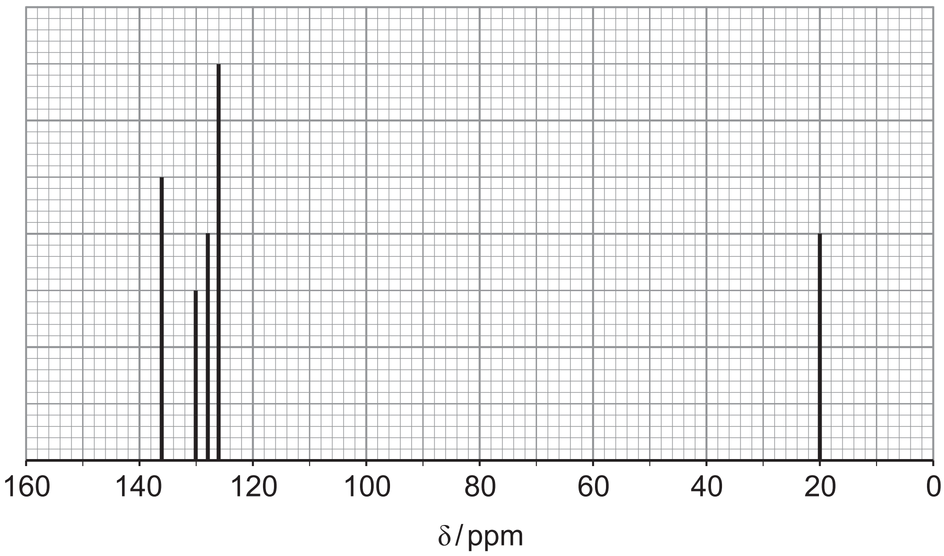
Examiner only

(iv) One of the raw materials used to make polyester **P** is methylbenzene. Under suitable conditions methylbenzene can be converted to a mixture of dimethylbenzene isomers and benzene.

$$2C_6H_5CH_3 \longrightarrow C_6H_4(CH_3)_2 + C_6H_6$$

Gas chromatography can be used to separate the dimethylbenzene isomers. One of the dimethylbenzene isomers has the ^{13}C NMR spectrum below.



Explain how this spectrum shows that the isomer is 1,3-dimethylbenzene. [3]

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(v) The isomer of dimethylbenzene used to produce polyester **P** is 1,4-dimethylbenzene. In industry this isomer is oxidised by air to produce benzene-1,4-dicarboxylic acid.

$$HOOC—C_6H_4—COOH$$

During this reaction a number of other aromatic oxidation products are made. One of these has the empirical formula C_4H_3O.

Suggest a structure for this product giving your reasoning. State how it could be produced in the reaction between 1,4-dimethylbenzene and atmospheric oxygen. [5]

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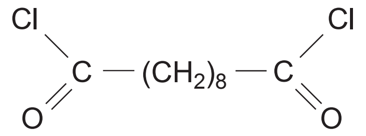
(b) (i) A laboratory experiment to demonstrate the formation of a polyamide uses decanedioyl dichloride.



State the name of a reagent used to produce this compound from the corresponding dicarboxylic acid, decanedioic acid. [1]

....

(ii) After use, a bottle containing decanedioyl dichloride (M_r 239.1) can become contaminated with decanedioic acid.

Analysis of a 3.50 g sample of contaminated decanedioyl dichloride showed that it contained 0.977 g of chlorine by mass.

Calculate the percentage purity of this sample of decanedioyl dichloride. [2]

Percentage purity = ... %

(iii) Suggest how the decanedioyl dichloride had become contaminated with decanedioic acid. [1]

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END OF PAPER

Surname	Centre Number	Candidate Number
First name(s)		2

GCE A LEVEL

1410U40-1

Z22-1410U40-1

MONDAY, 20 JUNE 2022 – MORNING

CHEMISTRY – A2 unit 4
Organic Chemistry and Analysis

1 hour 45 minutes

	For Examiner's use only		
	Question	**Maximum Mark**	**Mark Awarded**
Section A	**1. to 7.**	**10**	
Section B	**8.**	**14**	
	9.	**17**	
	10.	**13**	
	11.	**12**	
	12.	**14**	
	Total	**80**	

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01

ADDITIONAL MATERIALS

In addition to this examination paper, you will need a:
- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid. You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions.
Section B Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

Candidates are advised to allocate their time appropriately between **Section A (10 marks)** and **Section B (70 marks)**.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q.10**(a).

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SECTION A

Answer **all** questions.

1. A hydrocarbon is either cyclopentane or pent-1-ene.

Use the **Data Booklet** to choose a characteristic infrared absorption that will positively identify the compound as pent-1-ene. [1]

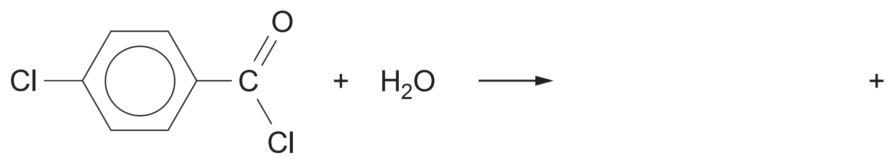
....

....

2. Give the **name** of a compound of formula C_4H_8 that will decolourise acidified potassium manganate(VII). [1]

....

3. Complete the equation for the reaction shown below. [1]



4. Give the structure of a compound of molecular formula $C_4H_8Br_2$ whose low resolution ^1H NMR spectrum consists of three peaks. [1]

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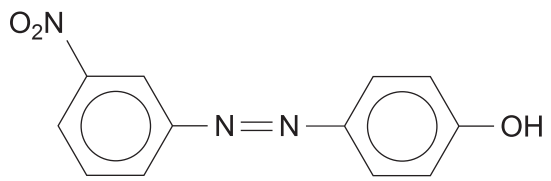
5. The formula of an azo dye, produced by reacting a diazonium compound with phenol, is shown below.



Write the formula of the amine that is used to produce this diazonium compound. [1]

6. Ethanal reacts with hydrogen cyanide to give 2-hydroxypropanenitrile, $CH_3CH(OH)(CN)$.

(a) State the type of mechanism occurring during this reaction. [1]

....

(b) Draw the structures of the two enantiomers of 2-hydroxypropanenitrile. [1]

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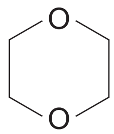
7. Three isomers have the molecular formula $C_4H_8O_2$.

(a) (i) Give the structure of an isomer that gives bubbles of carbon dioxide when tested with aqueous sodium hydrogencarbonate. [1]

(ii) Give the structure of an isomer that is both a ketone and a secondary alcohol. [1]

(b) Describe and explain the high resolution ^1H NMR spectrum of 1,4-dioxane.

The position of the signal(s) is **not** required. [1]



1,4-dioxane

....

....

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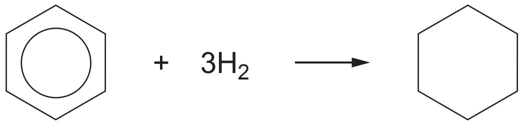
SECTION B

Answer **all** questions.

8. (a) The starting material for making polyamides is often benzene, obtained from petroleum.

(i) The first stage in the production of a particular polyamide is the hydrogenation of benzene at increased temperature and pressure.



Suggest the name of a catalyst that can be used for the hydrogenation of alkenes and an arene, such as benzene. [1]

....

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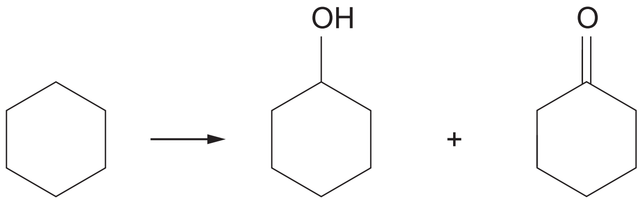
(ii) Cyclohexane, the product from the first stage, is then catalytically oxidised using air. This results in KA (a mixture of cyclohexanol and cyclohexanone). The mixture typically contains around 60% of cyclohexanone.



I. Using characteristic infrared values, describe how you could identify both cyclohexanol and cyclohexanone in KA. You should identify the bonds and their absorption frequencies in your answer. [2]

....

....

....

II. A student suggested that you could find the proportion of cyclohexanol and cyclohexanone in KA by comparing the percentage of oxygen in each compound.

Calculate the percentage of oxygen in each compound. Use your answers to explain why this method would probably not give an accurate answer for the proportion of each compound present in KA. [3]

....

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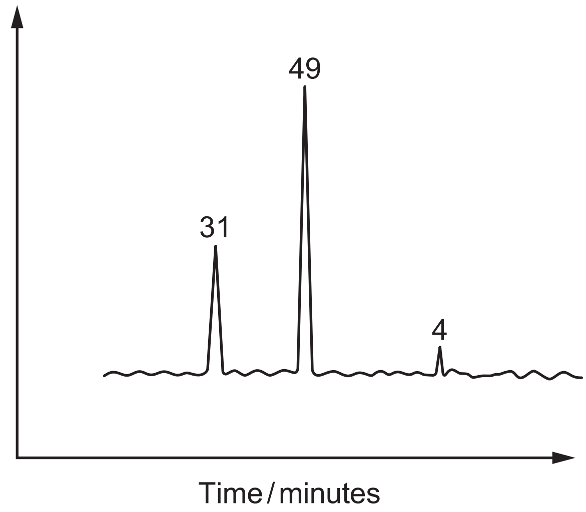
III. The proportions present in KA can be found more effectively by using gas chromatography.

A typical chromatogram of a sample of KA is shown below. The largest peak is due to cyclohexanone.



Calculate the percentage by volume of cyclohexanone in the mixture. [1]

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Percentage of cyclohexanone = ... %

(iii) A sample of the cyclohexanol in KA is oxidised in the laboratory by refluxing with a suitable reagent.

State a suitable oxidising agent that can be used. [1]

....

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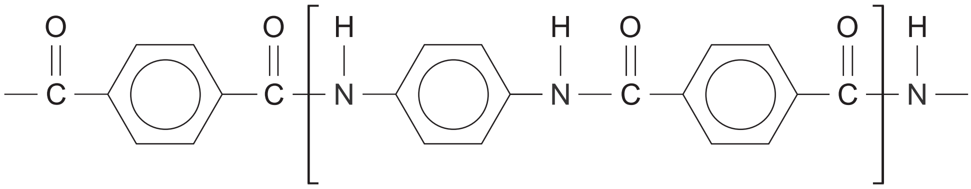
(b) The formula of the polyamide 'Kevlar' is shown below.



Give the structures of two starting materials that can react together to produce this polymer. [2]

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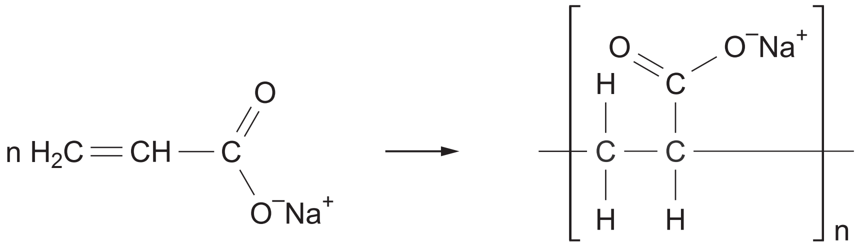
(c) The polymer poly(sodiumacrylate) is an example of a super absorbing polymer.



(i) Explain why this polymer is described as an addition polymer and not as a polyester. [2]

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....

(ii) Poly(sodiumacrylate) is capable of absorbing several hundred times its own mass of water, whilst remaining as a solid material.

In an experiment, 4.0×10^{-6} mol of the polymer absorbed 150 g of water. This mass of water is 300 times the mass of the polymer used.

Calculate the relative molecular mass of the polymer. [2]

M_r = ...

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09

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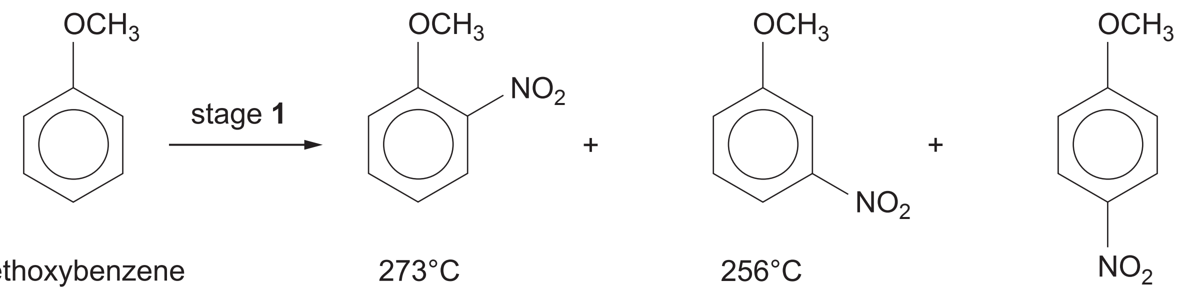
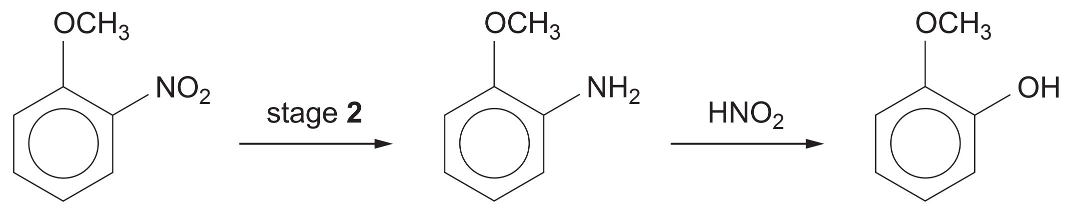
9. (a) Guaiacol (2-methoxyphenol) can be produced in a three-stage method from anisole (methoxybenzene). The boiling temperature of each product of stage **1** is shown.



methoxybenzene 273°C 256°C 260°C



guaiacol

Use your knowledge of a similar reaction, using benzene as the starting material, to answer parts (i)-(iii).

(i) Suggest a reagent(s) for stage **1**. [1]

....

(ii) Suggest a reagent(s) for stage **2**. [1]

....

(iii) Suggest and explain a practical problem that might occur at the end of stage **1**, before proceeding to stage **2**. [1]

....

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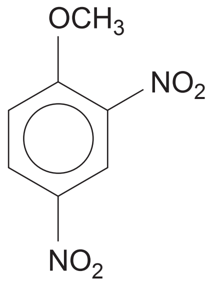
Examiner only

(iv) Complete the table below, indicating what is observed (if anything) when guaiacol reacts with aqueous solutions of these reagents. [2]

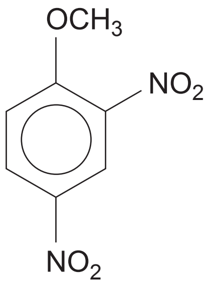
Reagent	$FeCl_3$	$NaHCO_3$
Observation		

(b) 2,4-Dinitroanisole can also be made by the nitration of anisole.



This compound has been used as an explosive. On detonation, it gives carbon, carbon monoxide, nitrogen and steam.

Balance the equation for this reaction. [1]



$\longrightarrow$ .... C + ... CO + ... N_2 + ... H_2O

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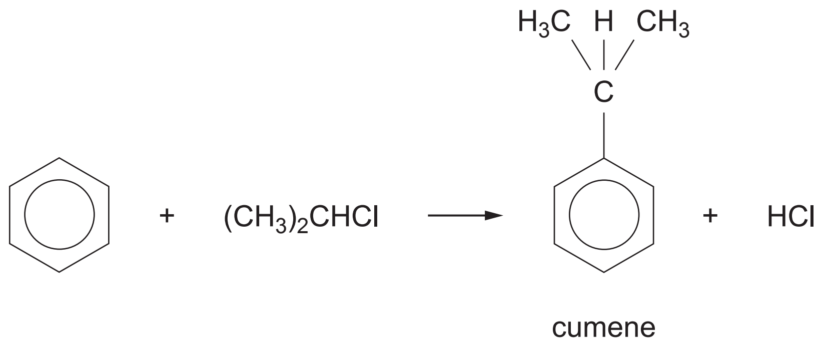
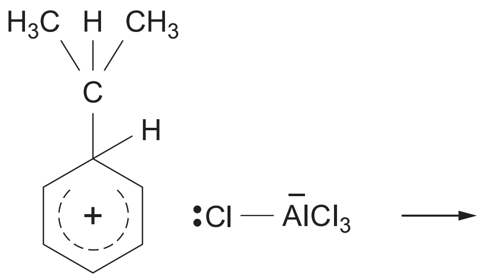
(c) (1-Methylethyl)benzene (commonly called cumene) can be prepared from benzene by a Friedel-Crafts reaction.



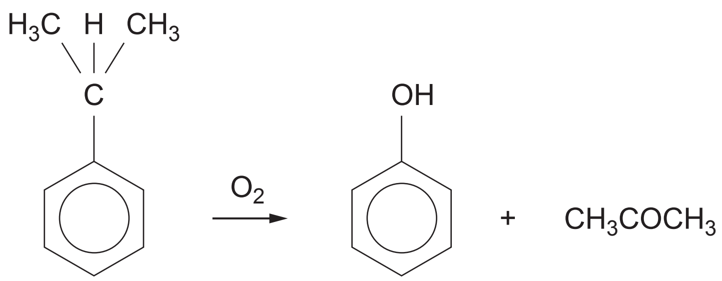
The products of the first step of the reaction are given below.

Complete the mechanism, showing how cumene, hydrogen chloride and the catalyst are formed from the intermediate product. [2]



Examiner only

(d) Cumene is an important material in the industrial preparation of phenol and propanone.



(i) The yield of phenol (M_r 94) in this process is 86% with benzene (M_r 78) as the starting material.

Calculate the mass of phenol produced from 234 kg of benzene. [2]

Mass of phenol = ... kg

(ii) During this reaction a cumene radical is produced.

State what is meant by the term 'radical'. [1]

....

....

(iii) Give the formula of a radical formed during the reaction of methane with chlorine. [1]

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(e) Phenol reacts with aqueous bromine to give 2,4,6-tribromophenol.

If aqueous bromine is added slowly to an aqueous solution of phenol, describe how you would know when the reaction is just complete. [1]

....

....

(f) Phenol reacts with benzoyl chloride, to give the ester phenyl benzoate and hydrogen chloride.

(i) Write the equation for this reaction. [1]

(ii) The reaction in part (i) is carried out in the presence of aqueous sodium hydroxide, which removes HCl as it is formed.

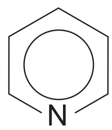
Explain why this method would not be suitable for a similar reaction, using ethanoyl chloride in place of benzoyl chloride. [1]

....

....

(iii) Esterification reactions, such as those in part (i), are sometimes carried out using pyridine in place of sodium hydroxide.

Explain the feature present in the structure of pyridine that enables it to react in this way. [2]



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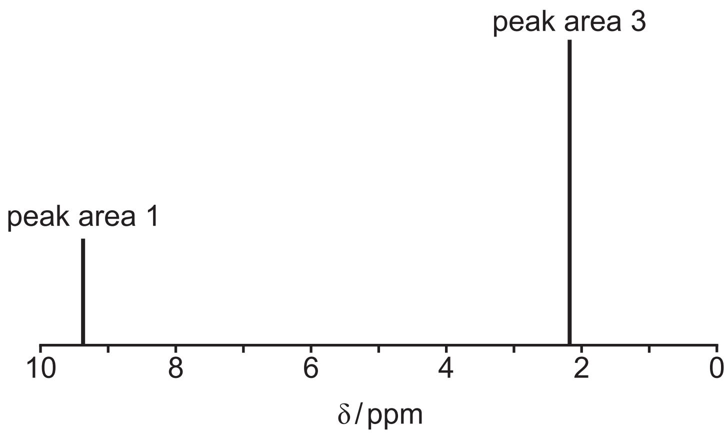
17

10. (a) Compound **G** contains only carbon, hydrogen and oxygen.

It has a molar mass of 72 g mol^{-1} of which 50.0% is carbon. The compound reacts positively with Tollens' reagent and gives a yellow solid when treated with alkaline iodine solution. It reacts with sodium tetrahydridoborate(III) to give a new compound which has a molar mass of 76 g mol^{-1}.

The high resolution ^{1}H NMR spectrum of compound **G** is shown below.



Examiner only

Use **all** of this information to deduce a structure for compound **G**.

You should comment on how each piece of data has helped you to deduce the structure. [6 QER]

Examiner only

(b) The absorption spectrum of compound **G** shows a maximum absorption at 450 nm.

Light energy measured in kJ mol^{-1} (E) is related to wavelength (λ) by the equation

$$E = \frac{\text{constant}}{\lambda}$$

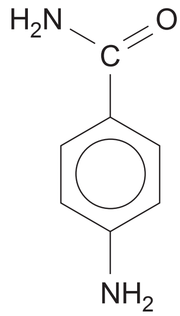
The energy of light of wavelength 656 nm is 183 kJ mol^{-1}.

Calculate the energy of the maximum absorption of compound **G** at 450 nm. [2]

Energy = ... kJ mol^{-1}

Examiner only

(c) The formula of 4-aminobenzamide is



(i) When this compound is warmed with excess aqueous sodium hydroxide, only the amide group reacts.

I. Give the structure of the organic product. [1]

II. State the nature of the attacking reagent and explain why the NH_2 group bonded to the benzene ring is not removed. [2]

....

....

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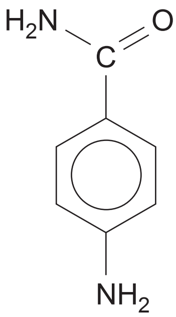
(ii) Both the functional groups in 4-aminobenzamide are attacked by nitric(III) acid at room temperature, giving 4-hydroxybenzoic acid, nitrogen gas and water as the products.

Complete the equation for this reaction. [2]

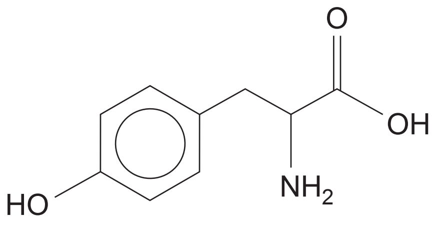


+ .... HNO_2 $\longrightarrow$ + ... N_2 + ... H_2O

13

Examiner only

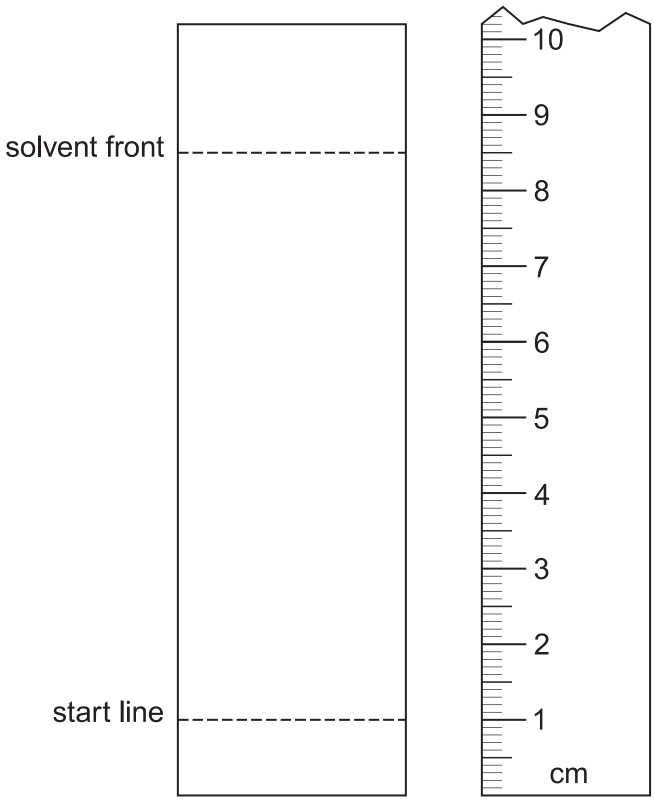
11. (a) Tyrosine is one of the amino acids making up casein, the protein in milk.



tyrosine

(i) Using a particular solvent tyrosine has an R_f value of 0.67.

On the chromatogram below show the spot given by tyrosine. [1]



Examiner only

(ii) Write the formula for the zwitterion of tyrosine. [1]

(iii) Tyrosine is described as a hydrophobic amino acid, as its solubility in water is very low.

Use the formula of tyrosine to give a reason for this low solubility. [1]

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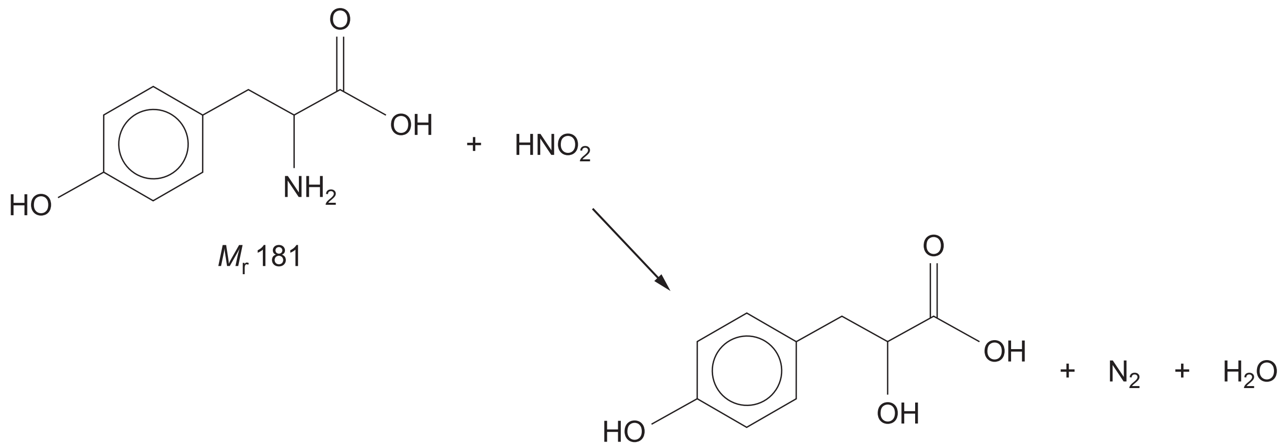
Examiner only

(iv) Amino acids, such as tyrosine, react with nitric(III) acid to give the corresponding hydroxy acid and nitrogen gas.



I. A sample of tyrosine was treated with an excess of nitric(III) acid and produced 147 cm^3 of nitrogen, measured at 298 K and 1 atm pressure.

Show that this volume of nitrogen will be produced from 1.09 g of tyrosine. [2]

II. The experiment was repeated with a sample of tyrosine from a different batch.
This time the same starting mass gave 132 cm^3 of nitrogen under the same conditions.

Suggest **one** reason for this different result. [1]

....

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Examiner only

(b) There is increasing interest in the production of important chemicals by biotechnology, rather than from the use of fossil fuels.

One of these compounds is butane-1,4-dioic acid, which can be made from glucose.

$$C_6H_{12}O_6 \xrightarrow{\text{anaerobic bacteria}} HOOC(CH_2)_2COOH + \text{other products}$$

glucose M_r 180 — butane-1,4-dioic acid M_r 118

In practice, a constant pH is maintained by the addition of calcium hydroxide.
As butane-1,4-dioate ions are produced they react with calcium hydroxide to give insoluble calcium butane-1,4-dioate, which is then filtered from the mixture.

$$C_6H_{12}O_6 + Ca(OH)_2 \longrightarrow (CH_2)_2(COO)_2Ca + \text{other products}$$

glucose M_r 180 — $Ca(OH)_2$ M_r 74 — calcium butane-1,4-dioate

(i) Calculate the atom economy for the production of calcium butane-1,4-dioate. [2]

Atom economy = ... %

Examiner only

(ii) In an experiment, 54.0 g of glucose produced 41.2 g of calcium butane-1,4-dioate.

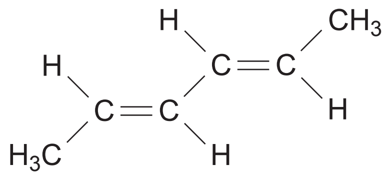
Calculate the minimum volume of sulfuric acid, of concentration 2.5 mol dm^{-3}, necessary to convert all the calcium butane-1,4-dioate into butane-1,4-dioic acid. [2]

$$(CH_2)_2(COO)_2Ca \quad + \quad H_2SO_4 \longrightarrow HOOC(CH_2)_2COOH \quad + \quad CaSO_4$$

Volume of sulfuric acid = ... cm^3

Examiner only

(c) Hexa-2,4-diene is one of the important products that can be produced from butane-1,4-dioic acid.



hexa-2,4-diene

Use the formula to help you to describe the ^{13}C NMR spectrum of this compound.

Your answer should include the number of peaks seen and your reasoning.
The position and size of the peaks is not required. [2]

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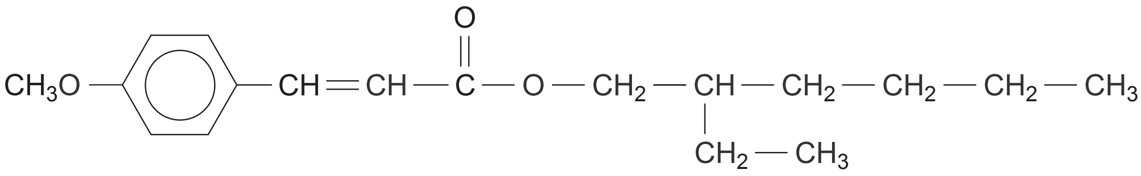
12

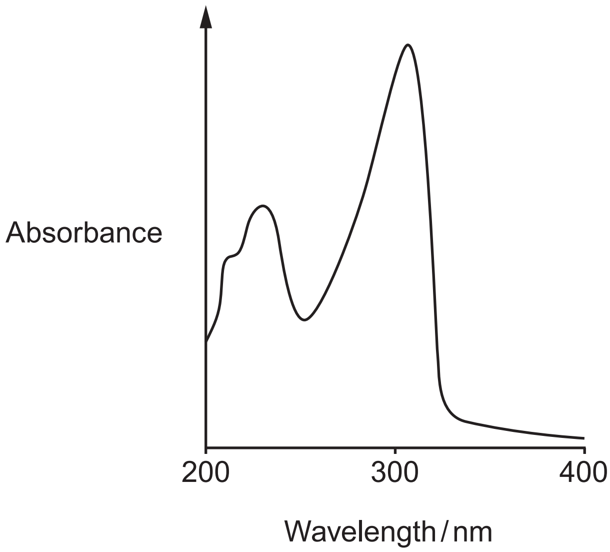
Examiner only

12. (a) People are increasingly concerned about the effect of the Sun's ultraviolet rays on the skin. A number of compounds have been developed for use as sunscreens.
One of these compounds is compound **A**.



compound **A**

(i) The ultraviolet absorption spectrum of compound **A** is shown below.



Ultraviolet radiation is divided into three regions.

Region	Wavelength / nm
UVA	320-400
UVB	290-320
UVC	200-290

Use this information to explain why it may be necessary to use another sunscreen compound, together with compound **A**, when producing a commercial sunscreen. [1]

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Examiner only

(ii) Compound **A** is an ester that is made by reacting together 2-ethylhexanol and the appropriate acid, in the presence of a catalyst.

Give the molecular formula of the acid used in this reaction. [1]

....

(iii) Compound **A** displays both forms of stereoisomerism.

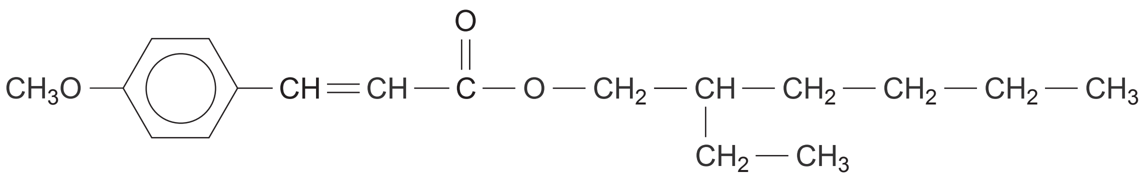
I. State how structural isomerism differs from stereoisomerism. [2]

....

....

....

II. Indicate the position of a chiral centre on the formula of compound **A** below. [1]



III. When compound **A** is made by the method described in part (ii) above, an equimolar mixture of both enantiomers is produced.

State how a solution of this mixture affects the plane of plane polarised light. [1]

....

IV. Exposure of a mixture of both *E*- and *Z*- forms of compound **A** to UV radiation results in the *E*- form gradually changing to the *Z*- form.

Using the general formula of compound **A** as R — CH ═ CH — R', draw and label these two forms of compound **A**. [2]

 (1410U40-1) **Turn over.**

Examiner only

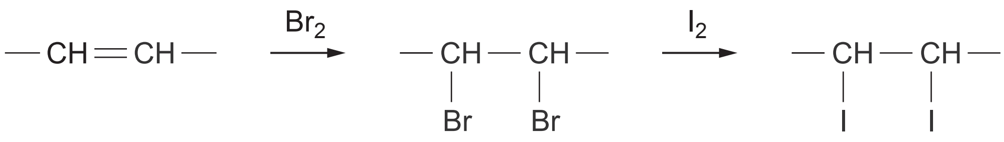
(b) Palm oil contains around 40% of unsaturated oil and 60% of saturated oil. The amount of unsaturation in an oil can be measured indirectly in a reaction with iodine.

(i) In the first stage of this method bromine adds across the C═C double bonds present.

State the type of mechanism occurring when bromine adds across these double bonds. [1]

....

(ii) The overall reaction with iodine can be represented by the following equation.



In palm oil, most of the unsaturated oils are present as glyceryl trioleate (M_r 885). This compound contains three C═C double bonds per molecule.

An 8.41 g sample of palm oil reacted indirectly with 0.0128 mol of iodine (I_2).

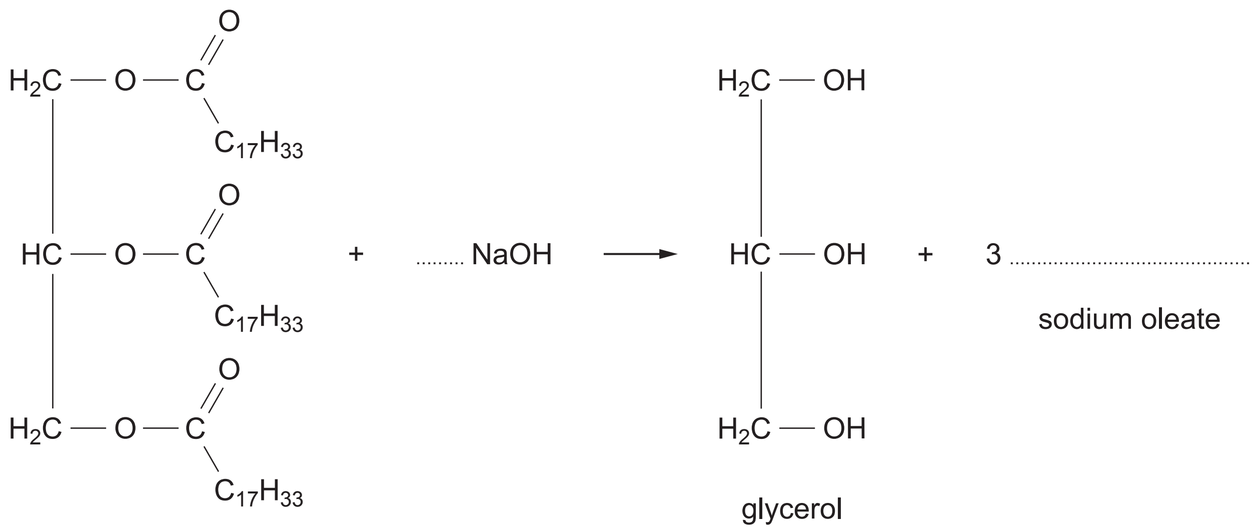
Calculate the percentage of unsaturated oil (as glyceryl trioleate) present in the palm oil. [2]

Percentage of unsaturated oil = ... %

Examiner only

(iii) Oils such as palm oil can be refined and used to make soaps such as sodium oleate.

Complete the equation below, which represents the alkaline hydrolysis of the unsaturated oil. [2]



(iv) Fats and oils are esters of carboxylic acids and glycerol.

Give the systematic name of glycerol (seen in part (iii) above). [1]

....

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END OF PAPER

Surname	Centre Number	Candidate Number
First name(s)		2

GCE A LEVEL

1410U40-1

S23-1410U40-1

MONDAY, 19 JUNE 2023 – AFTERNOON

CHEMISTRY – A2 unit 4

Organic Chemistry and Analysis

1 hour 45 minutes

	For Examiner's use only		
	Question	**Maximum Mark**	**Mark Awarded**
Section A	**1. to 7.**	**10**	
Section B	**8.**	**13**	
	9.	**15**	
	10.	**15**	
	11.	**15**	
	12.	**12**	
	Total	**80**	

1410U401 01

ADDITIONAL MATERIALS

In addition to this examination paper, you will need a:
- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions.
Section B Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

Candidates are advised to allocate their time appropriately between **Section A (10 marks)** and **Section B (70 marks)**.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q10**(a).

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Examiner only

SECTION A

Answer **all** questions.

1. Give the structure of an unsaturated aldehyde of molecular formula C_4H_6O. [1]

2. State a group that will give a positive triiodomethane (iodoform) test and give the observation for a positive result. [2]

....

3. (a) 1,2-Diaminoethane reacts as a base.

Explain how this compound acts as a base. [1]

....

....

(b) Give the structure of the organic compound formed when 1 mole of 1,2-diaminoethane reacts with 2 moles of ethanoyl chloride. [1]

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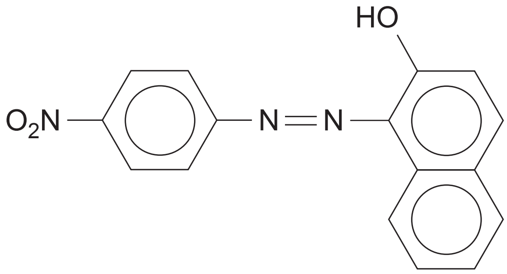
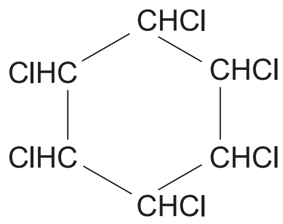
4. The formula of the azo dye, Para Red, is shown below.



Give the structure of the compound that couples with 4-nitrobenzenediazonium chloride to give this dye. [1]

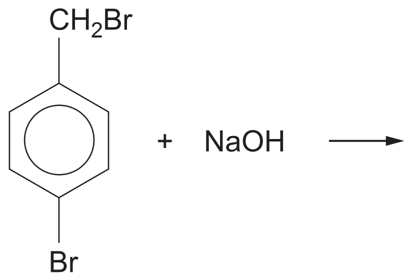
5. Hexachlorocyclohexane can be used as an insecticide.



Deduce the empirical formula of this compound. [1]

....

6. Give the structure of the organic product of the reaction below. [1]



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03

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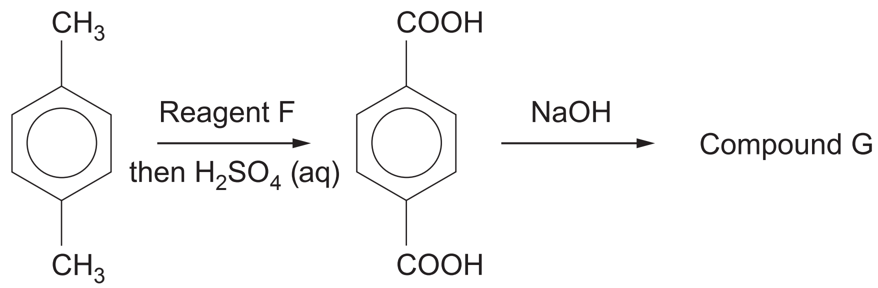
7. Study the reaction sequence below and answer the **two** questions that follow.



(a) Reagent F is .... [1]

(b) Give the structure of Compound G. [1]

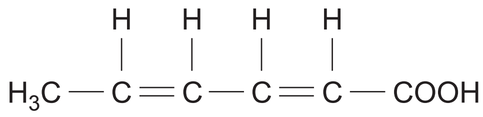
10

Examiner only

Section B

Answer **all** questions.

8. (a) Sorbic acid was isolated in 1859 from mountain ash berry oil. It is a white solid that is slightly soluble in cold water.



(i) The solubility of sorbic acid in water is 1.6 g dm^{-3} at 20 °C and 40.0 g dm^{-3} at 100 °C.

Calculate how much sorbic acid is precipitated from its aqueous solution if 200 cm^3 of a saturated solution at 100 °C is cooled to 20 °C.
Give your answer to an appropriate number of significant figures. [2]

.... g

1410U401 05

(ii) Sorbic acid and its salts, for example sodium sorbate, have important uses as antimicrobial agents in food preservation.
Some moulds are, however, able to detoxify the action of these sorbates.
An example is the decarboxylation of sodium sorbate.

I. State what is meant by 'decarboxylation'. [1]

....

II. In the laboratory sodium sorbate can be decarboxylated by heating it with soda lime.
The organic product of decarboxylation is *E*-penta-1,3-diene.

Write the equation for this decarboxylation of sodium sorbate with soda lime (which you should represent as NaOH in your equation), showing the structure of *E*-penta-1,3-diene. [2]

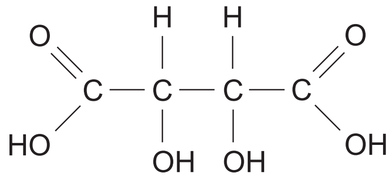
Examiner only

III. *E*-penta-1,3-diene can then react with hydrogen using platinum as a catalyst giving pentane.

State the number of moles of hydrogen required to react with 0.2 mol of *E*-penta-1,3-diene in this way. [1]

....

(b) Tartaric acid (2,3-dihydroxybutanedioic acid) is used in the food industry.



(i) Indicate any chiral centre(s) present by means of an asterisk (*). [1]

(ii) This acid occurs in a number of optically active forms.
Complete the sentences below.

Forms of tartaric acid that rotate the plane of plane-polarised light are called

.... [1]

A solution containing an equimolar mixture of two forms of the acid that rotates the plane of plane-polarised light in equal and opposite directions is called a

.... [1]

(c) Both sorbic acid and tartaric acid are described as showing stereoisomerism.

Explain what is meant by the term stereoisomerism. [1]

....

....

(d) Tartaric acid is formed when butenedioic acid reacts with a suitable oxidising agent.

Give the equation for this reaction, representing the oxidising agent as [O] and using water as a reactant. [1]

Examiner only

(e) The solubility of unsaturated carboxylic acids in water decreases as the chain length increases.

Number of carbon atoms in the alkyl chain	Solubility in water at 25 °C / g dm^{-3}
3	94
5	1.6
7	0.7

Suggest why this solubility decreases as the chain length increases. [2]

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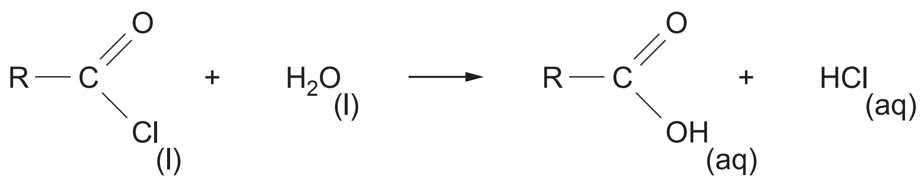
1410U401 07

13

Examiner only

9. (a) 1.598 g of an acid chloride, RCOCl, was added to water.
The aqueous solution contained hydrochloric acid and a carboxylic acid, R — COOH, where R is an alkyl group



(i) State the type of reaction occurring. [1]

....

(ii) This acidic solution was titrated with 0.400 mol dm^{-3} aqueous sodium hydroxide, using a suitable indicator. Both acids were just neutralised by 75.00 cm^{3} of the sodium hydroxide solution.

Use the results to calculate the relative molecular mass of the acid chloride. [3]

M_r ....

(iii) The low resolution ^{1}H NMR spectrum of the acid chloride showed two signals in the peak area ratio of 6:1.

Use this information and the relative molecular mass of the RCOCl, obtained in part (ii), to find the structure of the acid chloride. [2]

Examiner only

(b) (i) The acid chloride, benzene-1,4-dicarbonyl dichloride, ClOC — C_6H_4 — COCl is made by reacting benzene-1,4-dicarboxylic acid with phosphorus(V) chloride. The other products of this reaction are hydrogen chloride and phosphoryl trichloride, $POCl_3$.

Give the equation for this reaction. [1]

(ii) Benzene-1,4-dicarbonyl dichloride reacts with benzene-1,4-diamine to give a polyamide.

Show the repeating unit for this polyamide. [1]

1410U401
09

(c) Nylon polyamides that are produced from starting materials with a different number of carbon atoms are given numbers.
For example Nylon 4,5 has a 4-carbon diamine fragment and a 5-carbon dicarboxylic acid fragment.

Butane-1,4-diamine can be used as a starting material for this polyamide. This can be produced in a two-stage reaction from butane-1,4-diol.

$$HOCH_2(CH_2)_2CH_2OH \xrightarrow[\text{Stage 1}]{\text{reagent } \mathbf{A}} ClCH_2(CH_2)_2CH_2Cl \xrightarrow[\text{Stage 2}]{\text{reagent } \mathbf{B}} H_2N(CH_2)_4NH_2$$

(i) State the name of reagent(s) **A**. [1]

....

(ii) State the name of reagent(s) **B**. [1]

....

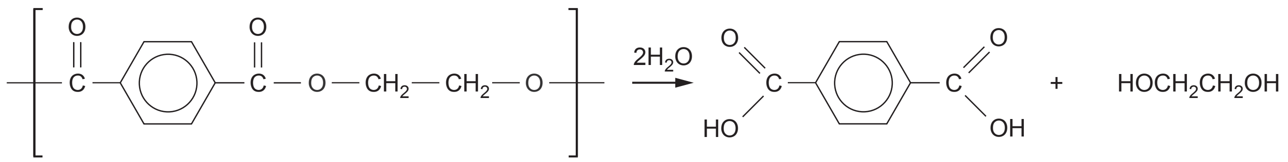
Examiner only

(iii) State the type of mechanism occurring in Stage 2. [1]

....

(iv) Draw the **skeletal** formula of the 5-carbon containing dicarboxylic acid used to produce Nylon 4,5. [1]

(d) The depolymerisation of polyamides and polyesters present a number of difficult problems as these polymers are very stable and only slowly decompose in the environment.
Poly(ethyleneterephthalate) (PET) is very difficult to hydrolyse but a new process using specific enzymes is proving promising. In this process 90% of PET is depolymerised into benzene-1,4-dicarboxylic acid.



Calculate the mass of benzene-1,4-dicarboxylic acid produced from 75 kg of PET if the yield from this hydrolysis is 90%. [3]

mass of benzene-1,4-dicarboxylic acid produced = ... kg

15

 (1410U40-1)

Examiner only

10. (a) Benzene reacts with bromine in the presence of a catalyst.
Give the mechanism for this reaction and explain how the Br — Br bond becomes polarised during the reaction.
Suggest why, in the absence of this catalyst, there is very little reaction between benzene and bromine under normal conditions. [6 QER]

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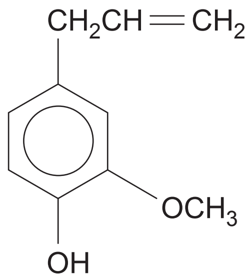
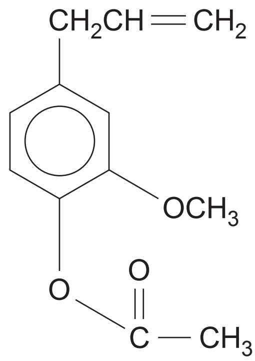
11

Examiner only

(b) Eugenol is the main constituent of clove oil, together with smaller quantities of eugenyl ethanoate.



Eugenol



Eugenyl ethanoate

(i) Describe what is seen if a few drops of iron(III) chloride solution are added to a solution of eugenol. [1]

....

(ii) State the colour change that occurs if a few drops of aqueous bromine are added to a solution of eugenol. [1]

....

(iii) If an excess of bromine is added to eugenol, a new compound is formed that contains the following percentages by mass of each element.

C 24.9% H 2.1% O 6.6%, the remainder being the % of bromine

Use this information to calculate the empirical and molecular formulae of this brominated compound and suggest a possible structure for it. [5]

Examiner only

(iv) Explain how adding aqueous sodium hydroxide at room temperature to a solution of eugenol and eugenyl ethanoate dissolved in trichloromethane, enables the two compounds to be separated.
Trichloromethane and water are immiscible. [2]

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Examiner only

11. (a) Aminoethanoic acid is the simplest α-amino acid.

$$H_2N — CH_2 — COOH$$

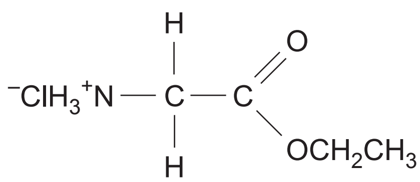
(i) Give the structure of the dipeptide formed from two molecules of aminoethanoic acid. [1]

(ii) Explain why aminoethanoic acid can only form one dipeptide. [1]

....

....

(b) Amino acids form esters in the usual way from the carboxylic acid group and an alcohol, in the presence of an acid catalyst. These esters can form a salt with the acid used in esterification.
The formula of one of these salts is shown below. Compound **M** is a white solid that is soluble in water, giving a colourless solution.



Compound **M**

(i) I. Many amino acids exist as zwitterion forms in aqueous solution.

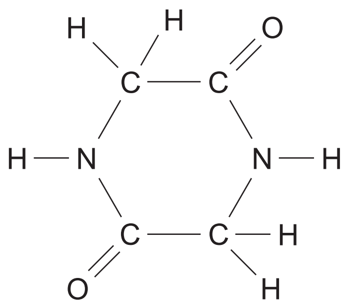
Give the zwitterion form of aminoethanoic acid. [1]

II. Explain why compound **M** cannot form a zwitterion in this way. [1]

....

Examiner only

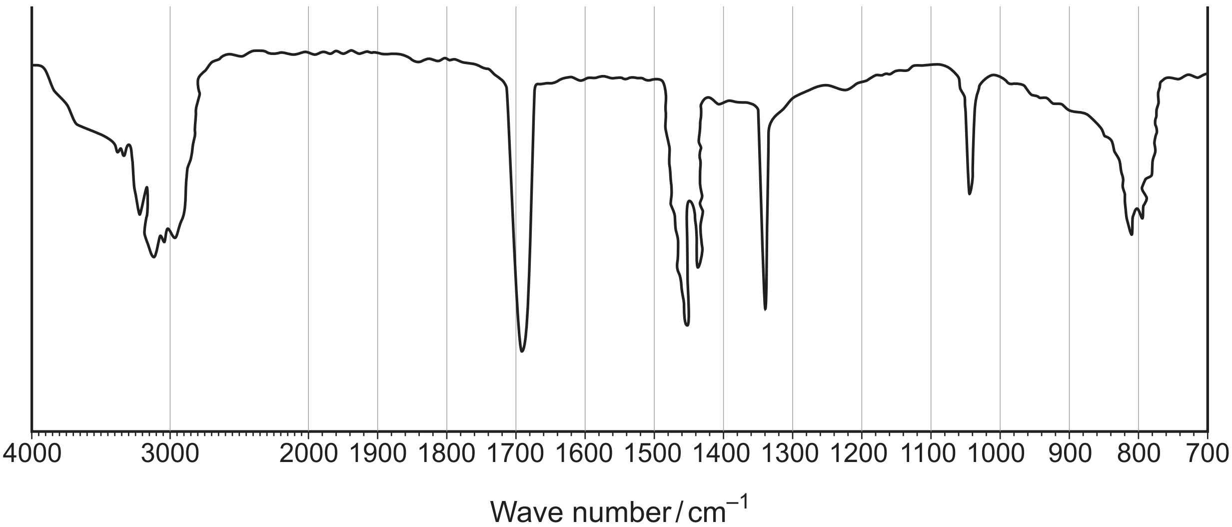
(ii) Under suitable conditions, aminoethanoic acid condenses to give diketopiperazine.



I. The mass spectrum of diketopiperazine shows a molecular ion at m/z 114 and two prominent fragments at m/z 43 and 71.

Suggest a formula for the fragment at m/z 71. [1]

II. An outline infrared absorption spectrum of diketopiperazine is shown below. Use the formula shown and the data sheet to suggest an absorption value for the N — H and C ═ O bonds. [1]



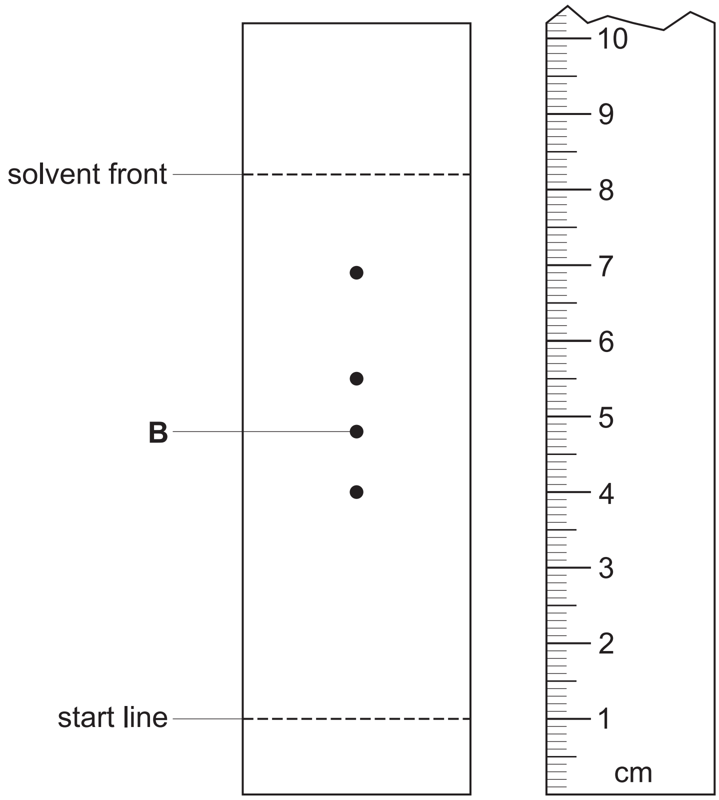
N — H .... cm^{-1} C ═ O .... cm^{-1}

Examiner only

(c) A mixture of α-amino acids can be separated and identified by thin layer chromatography.

(i) These amino acids are colourless and the chromatogram is sprayed with a solution of ninhydrin, so that the amino acids appear as purple dots. The colour is due to the dye Ruhemann's Purple.

A thin layer chromatogram of a mixture of α-amino acids is shown below.



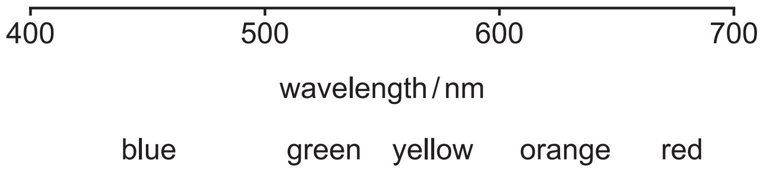
Calculate the R_f value for threonine (spot **B**). [1]

....

Examiner only

(ii) Ruhemann's Purple has a maximum absorption at 564 nm.

The information below indicates the colours absorbed at various wavelengths in the visible spectrum.
Use this to help you explain why the colour of the dye is purple. [1]



....

....

....

(iii) Calculate the energy (in kJ mol^{-1}) associated with this absorption at 564 nm. [3]

energy = ... kJ mol^{-1}

Examiner only

(d) Amino acids react with nitric(III) acid to produce nitrogen gas.

$$R—CH(NH_2)—COOH \;+\; HNO_2 \longrightarrow R—CH(OH)—COOH \;+\; N_2 \;+\; H_2O$$

(i) 0.500 g of 2-aminohexanoic acid, $CH_3(CH_2)_3CH(NH_2)COOH$, reacted with an excess of nitric(III) acid.

Calculate the volume of nitrogen produced, assume the temperature was measured at 298 K and at 1 atmosphere pressure. [2]

volume of nitrogen produced = ... cm^3

(ii) The actual volume of nitrogen produced was 90.9 cm^3, which was less than the calculated volume in part (i) above.

Suggest **two** reasons for this low result, apart from errors in weighing and in measuring the volume of nitrogen produced. [2]

1. ....

....

2. ....

....

15

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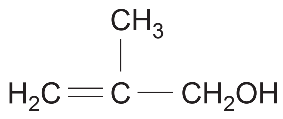
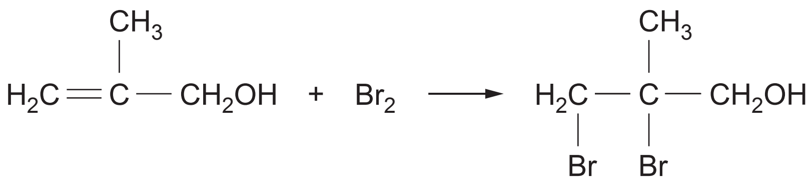
12. (a) 2-Methylpropenoic acid can be obtained from 2-methylprop-2-en-1-ol.



The standard reagents used for the oxidation of a primary alcohol to a carboxylic acid may affect the C$=$C bond. To prevent this occurring the double bond is protected by bromination.

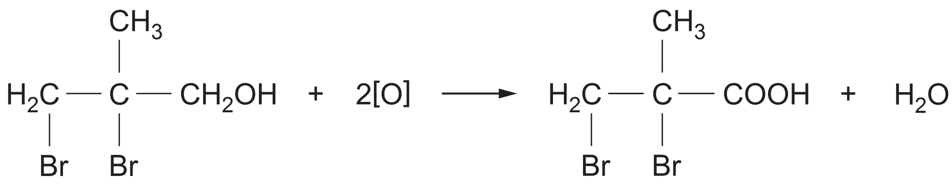


The bromine atoms are removed in a later reaction to give the required acid.

(i) State the type of mechanism occurring during the bromination. [1]

....

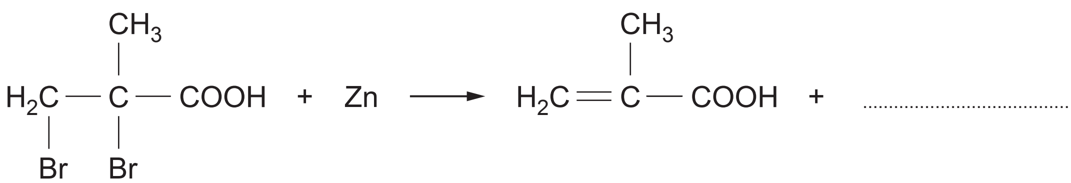
(ii) Suggest an oxidising agent used for the oxidation of the brominated alcohol. [1]



....

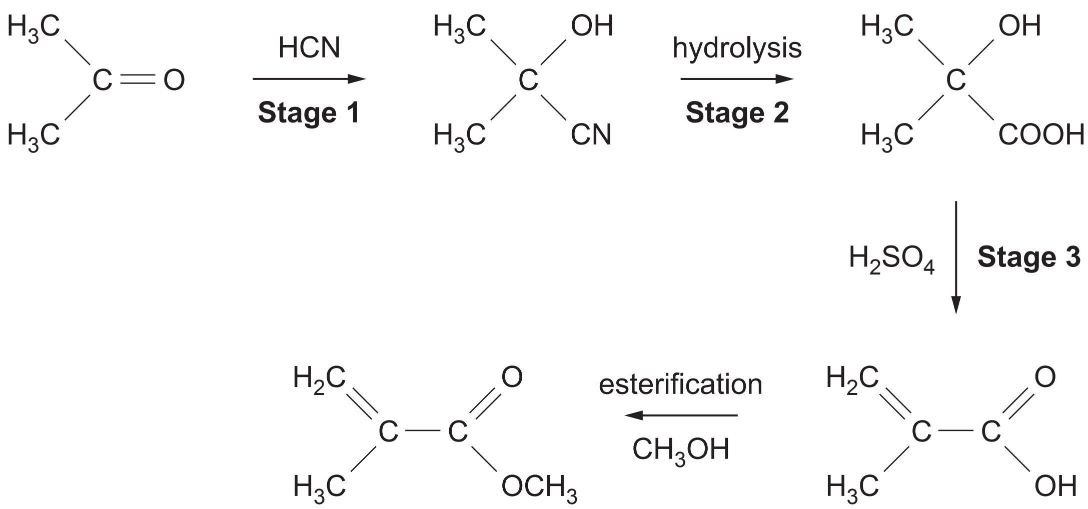
(iii) The dibromoacid produced in part (ii) is then reacted with zinc under suitable conditions to give 2-methylpropenoic acid.

Complete the equation, showing zinc bromide as the co-product. [1]



Examiner only

(b) Methyl 2-methylpropenoate can be produced from propanone.



(i) **Stage 1** is a nucleophilic addition reaction.

Give the formula of the nucleophile taking part in this stage. [1]

....

(ii) State why **Stage 1** is described as an **addition** reaction. [1]

....

(iii) State a reagent used for hydrolysis in **Stage 2**. [1]

....

(iv) State the role of sulfuric acid in **Stage 3**. [1]

....

(v) Addition polymerisation of methyl 2-methylpropenoate gives ‘Perspex’.

Give the repeating unit of this polymer. [1]

Examiner only

(vi) Explain why the polymerisation in part (v) is described as addition polymerisation and not condensation polymerisation. [1]

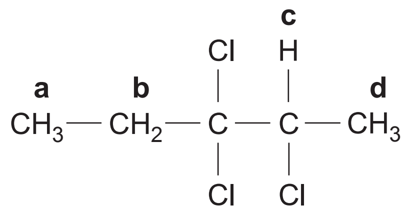
....

....

....

....

(c) The formula of a halogenoalkane is shown below.



(i) Give the systematic name of this halogenoalkane. [1]

....

(ii) Complete the table below which describes the high resolution ^{1}H NMR spectrum of this compound. [2]

Hydrogen proton	Splitting pattern
a	
b	
c	
d	

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END OF PAPER